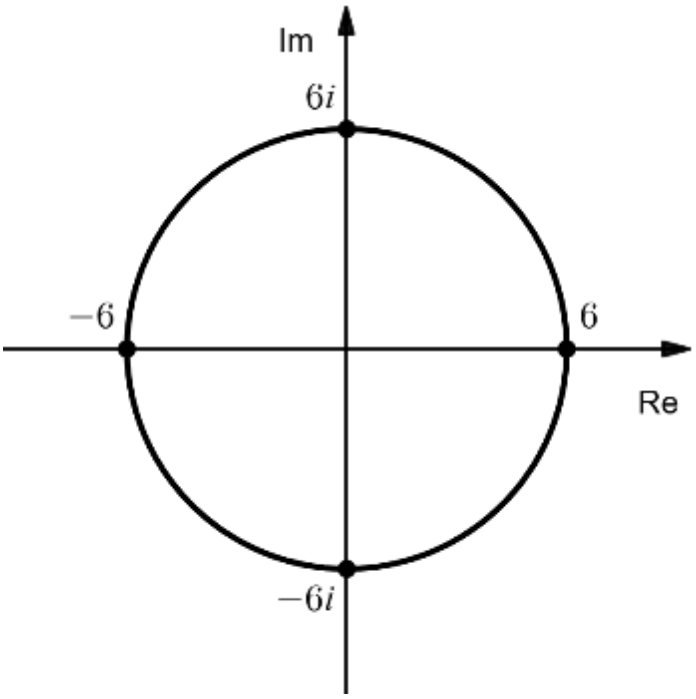
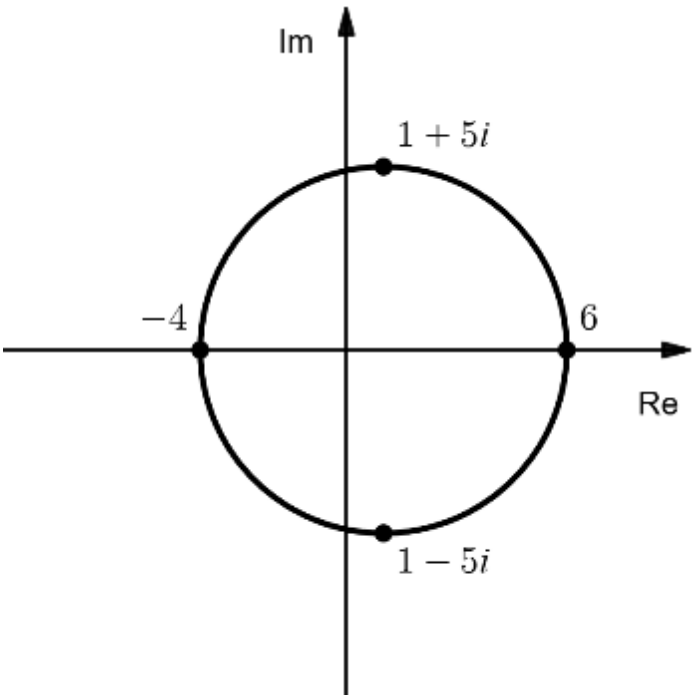
1.07 Complex Numbers: Loci in the Argand Diagram

Worked Solutions

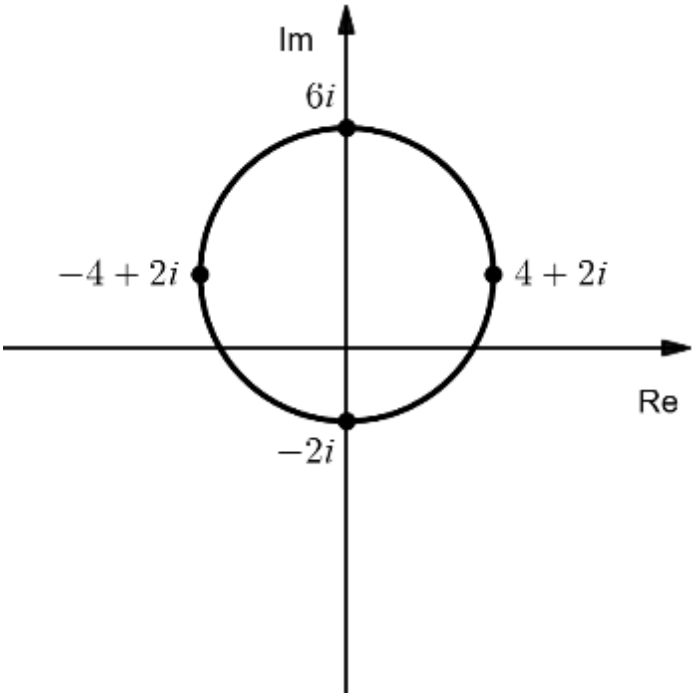
1 a)



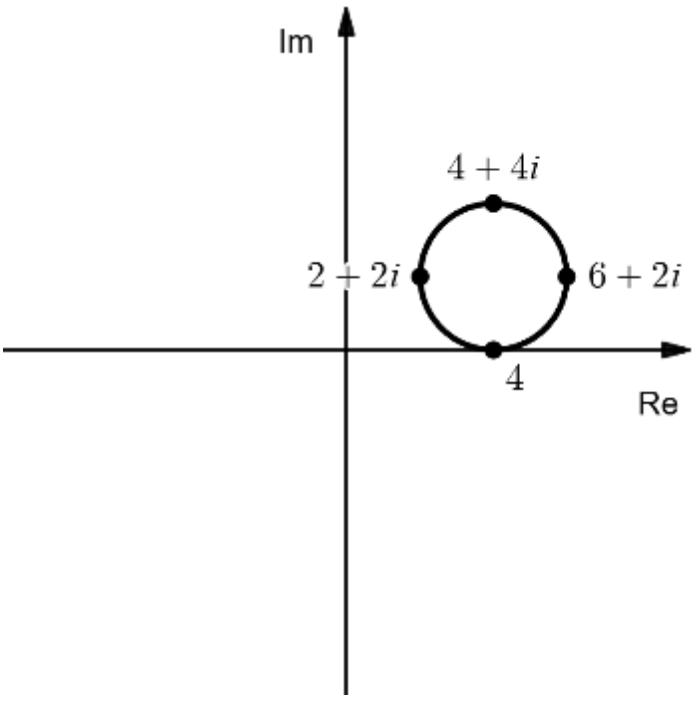
1 b)



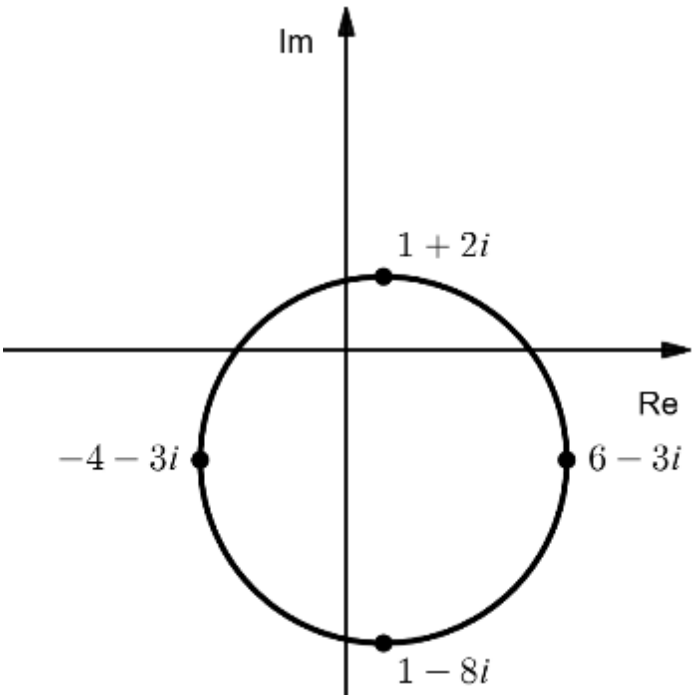
1 c)



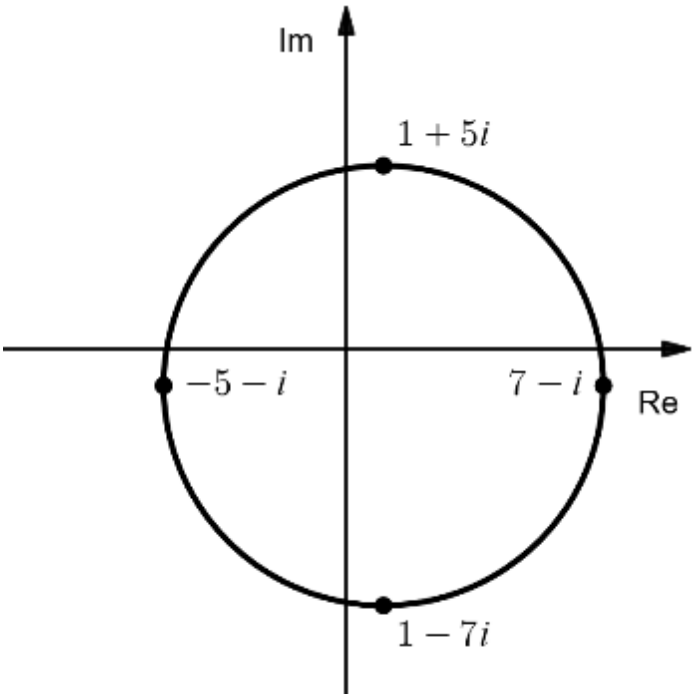
1 d)



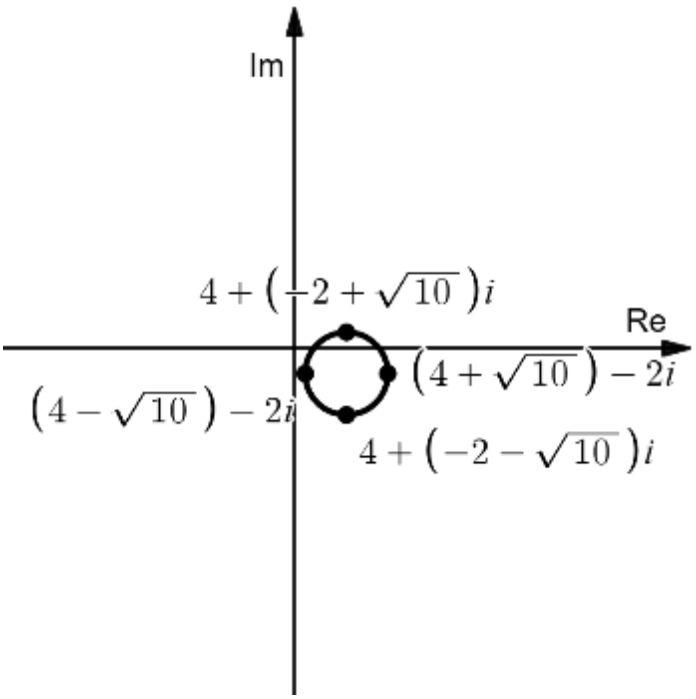
1 e)



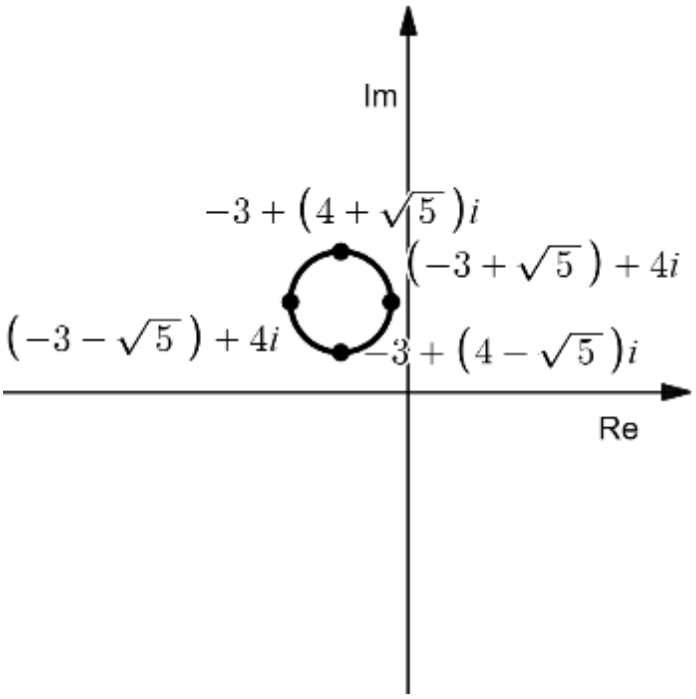
1 f)



1 g)



1 h)



2 a)

$$|z| = 4$$

2 b)

$$|z - 3i| = 5$$

2 c)

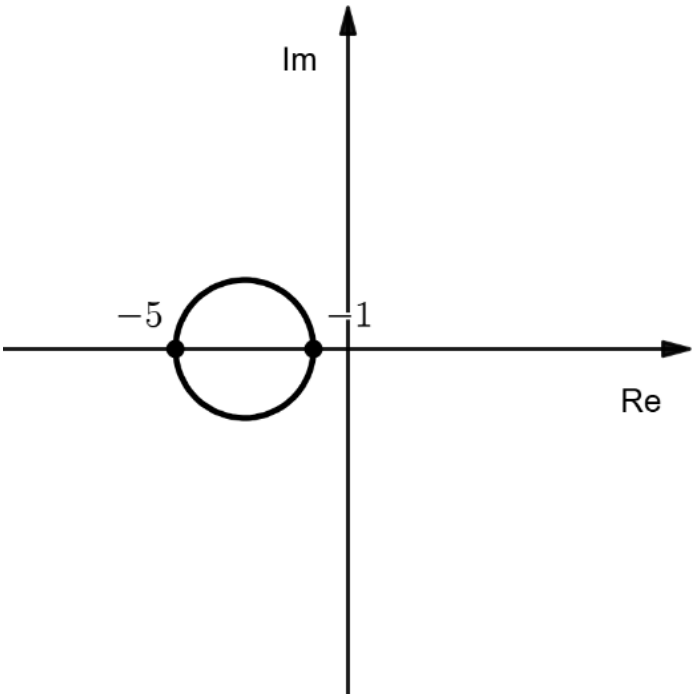
$$|z - 2 + 5i| = 2$$

2 d)

$$|z + 3 - 2i| = 4$$

3 a)

Centre is $(-3, 0)$ and radius is 2, so $(x + 3)^2 + y^2 = 4$

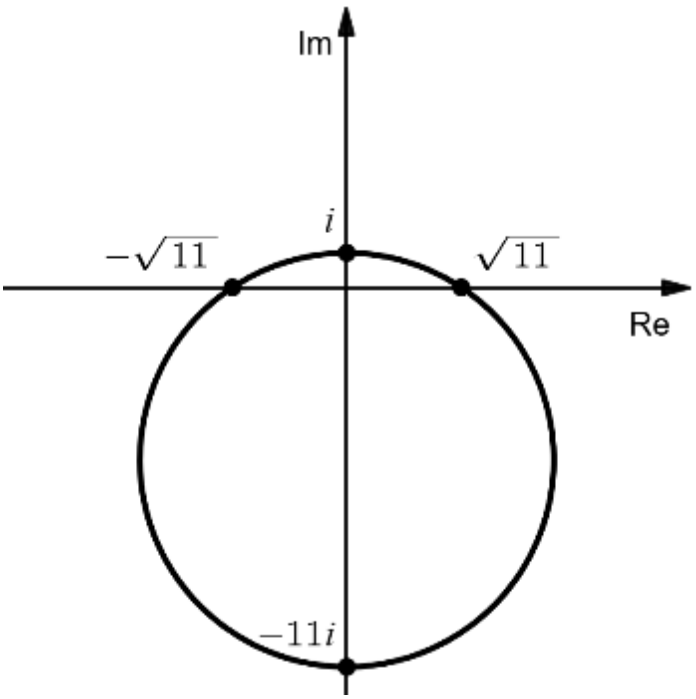


3 b)

Centre is $(0, -5)$ and radius is 6 so $x^2 + (y + 5)^2 = 36$

When $x = 0$, $y = -5 \pm \sqrt{36}$ so $y = 1$ or $y = -11$

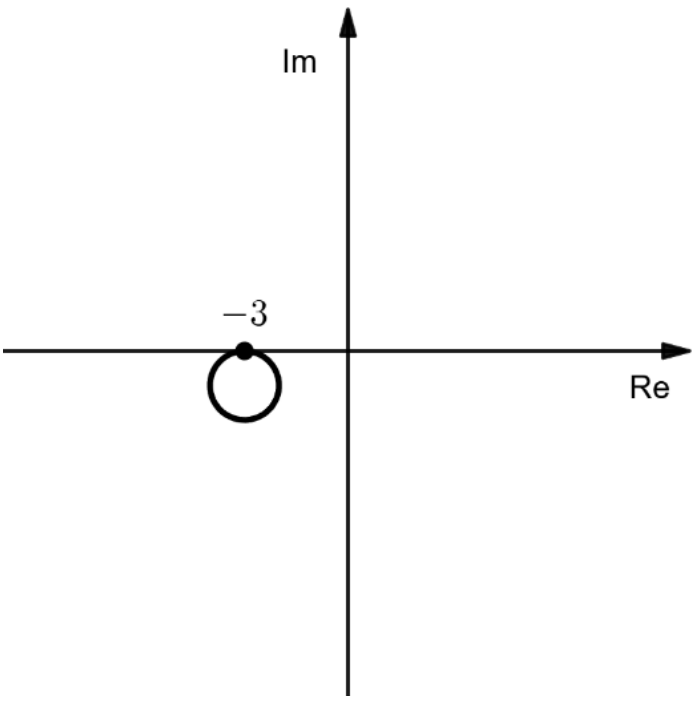
When $y = 0$, $x = \pm\sqrt{11}$



3 c)

Centre is $(-3, -1)$ and radius is 1 so $(x+3)^2 + (y+1)^2 = 1$

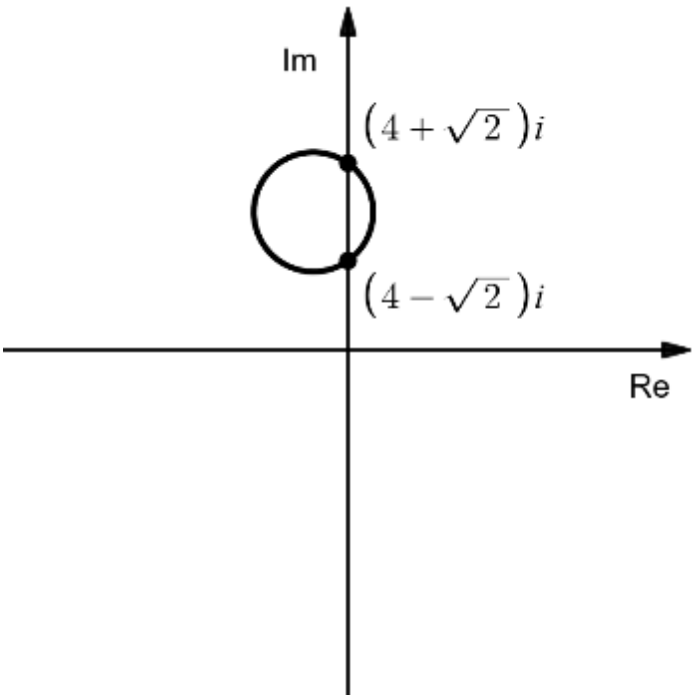
When $x = 0$, y has no real solutions, when $y = 0$, $x = -3$



3 d)

Centre is $(-1, 4)$ and radius is $\sqrt{3}$ so $(x+1)^2 + (y-4)^2 = 3$

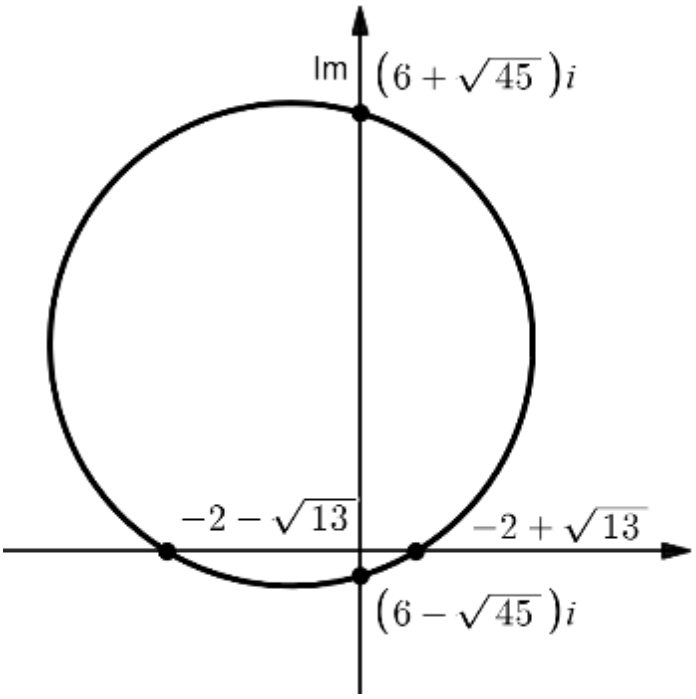
When $x = 0$, $y = 4 \pm \sqrt{2}$ and when $y = 0$, $x = \pm\sqrt{11}$



3 e)

Centre is $(-2, 6)$ and radius is 7 so $(x+2)^2 + (y-6)^2 = 49$

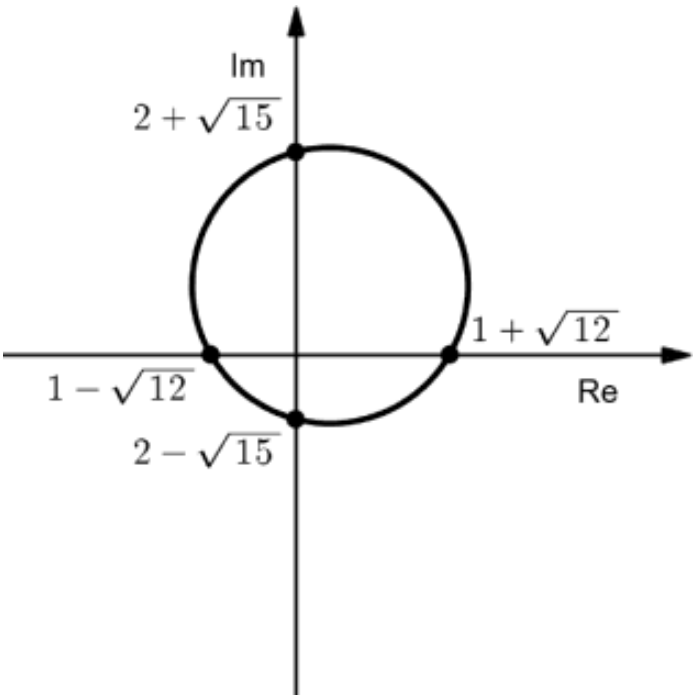
When $x = 0$, $y = 6 \pm \sqrt{45}$ and when $y = 0$, $x = -2 \pm \sqrt{13}$



3 f)

Centre is $(1, 2)$ and radius is 4 so $(x-1)^2 + (y-2)^2 = 16$

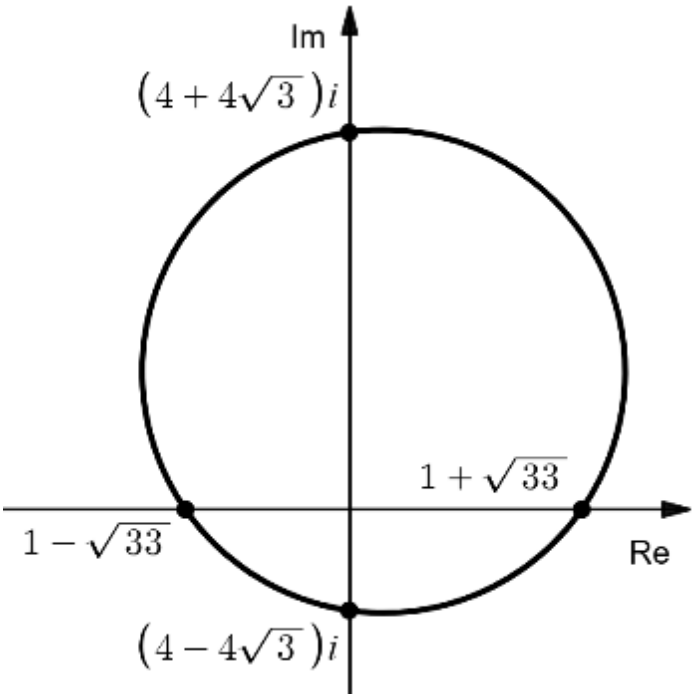
When $x = 0$, $y = 2 \pm \sqrt{15}$ and when $y = 0$, $x = 1 \pm \sqrt{12}$



4 a)

Centre is $(1, 4)$ and radius is 7 so $(x-1)^2+(y-4)^2=49$

When $x=0$, $y=4\pm4\sqrt{3}$ and when $y=0$, $x=1\pm\sqrt{33}$



4 b)

$|z-1-4i|=7$ and $\mathrm{Re}(z)=0$ when $z=(4\pm4\sqrt{3})i$

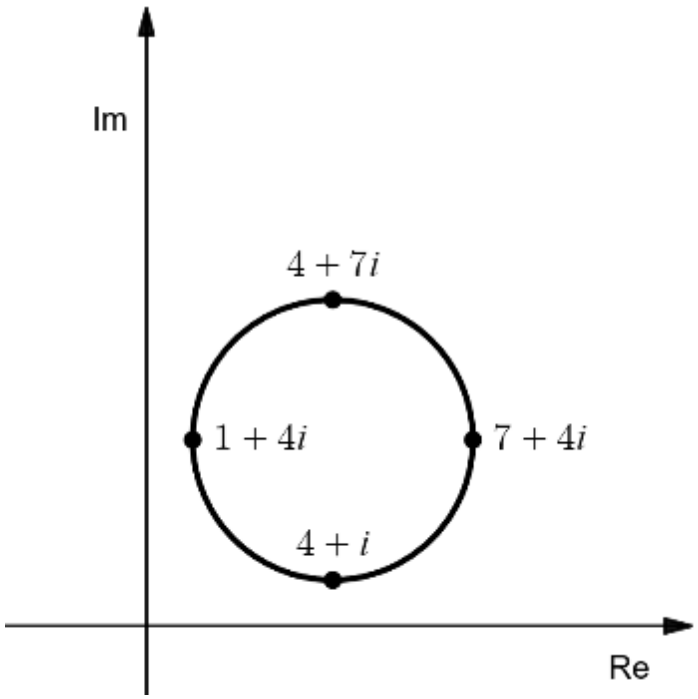
4 c)

$|z - 1 - 4i| = 7$ and $\text{Im}(z) = 0$ when $z = 1 \pm \sqrt{33}i$

5 a)

Centre is $(4, 4)$ and radius is 3 so $(x - 4)^2 + (y - 4)^2 = 9$

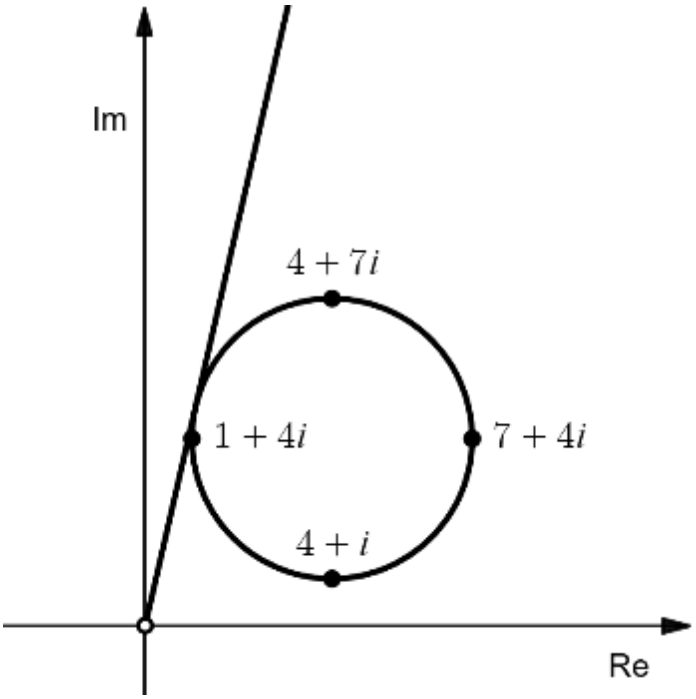
Since the radius is 3, the circle does not intersect with the axes, so fully lies within the top-right quadrant.

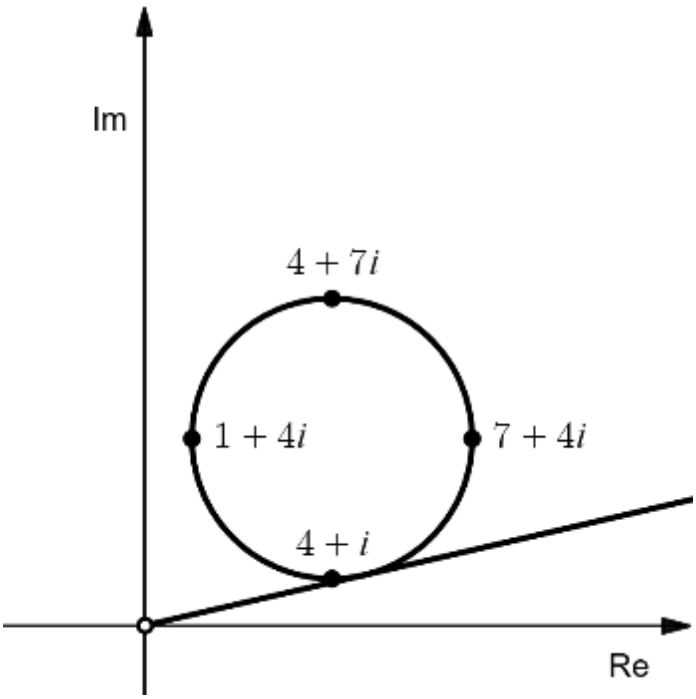


5 b)

$$(x - 4)^2 + (y - 4)^2 = 9$$

5 c)





Need to find when $y = mx$ and $(x-4)^2 + (y-4)^2 = 9$ have one point of intersection.

$$(x-4)^2 + (mx-4)^2 = 9$$
$$x^2 - 8x + 16 + m^2x^2 - 8mx + 16 - 9 = 0$$
$$(m^2+1)x^2 - 8(1+m)x + 23 = 0$$

A single point of intersection occurs when $b^2 - 4ac = 0$

$$64(1+m)^2 - 92(m^2+1) = 0$$
$$64m^2 + 128m + 64 - 92m^2 - 92 = 0$$
$$-28m^2 + 128m - 28 = 0$$
$$\therefore m = \frac{16 \pm 3\sqrt{23}}{7}$$

So the maximum $\arg z = \arctan\left(\frac{16+3\sqrt{23}}{7}\right) = 1.34$

and minimum $\arg z = \arctan\left(\frac{16-3\sqrt{23}}{7}\right) = 0.226$

5 d)

The maximum and minimum values of $|z|$ can be found by finding the distance from the origin to the centre, then adding or subtracting the radius:

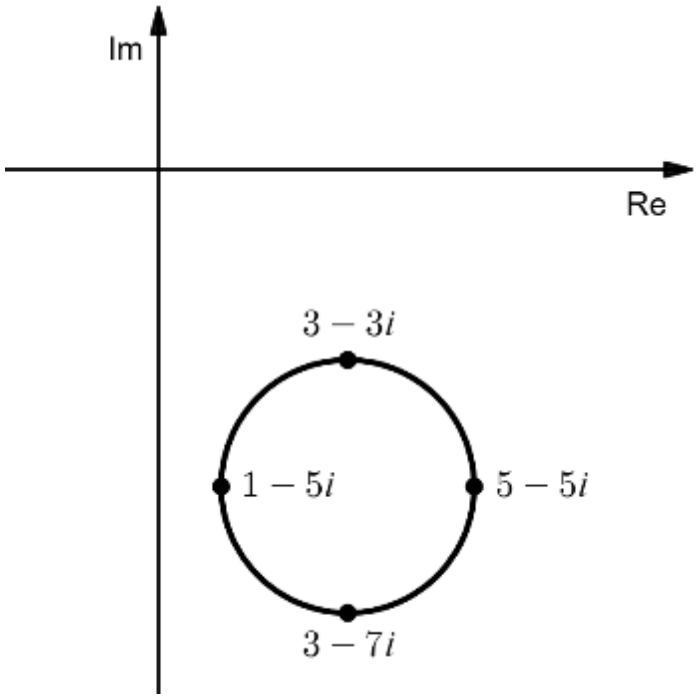
Maximum $|z| = 3 + \sqrt{4^2 + 4^2} = 3 + 4\sqrt{2}$

Minimum $|z| = 3 - \sqrt{4^2 + 4^2} = 3 - 4\sqrt{2}$

6 a)

Centre is $(3, -5)$ and radius is 2 so $(x-3)^2 + (y+5)^2 = 4$

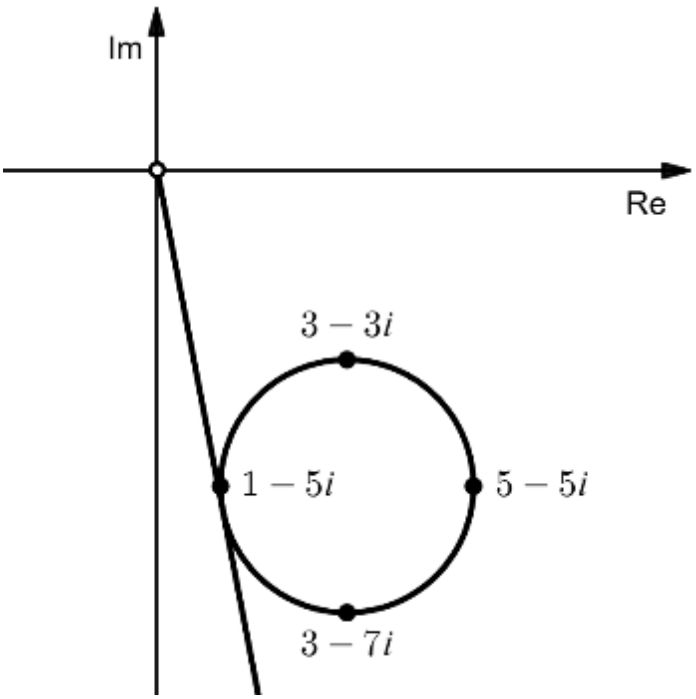
Since the radius is 2, the circle does not intersect with the axes, so fully lies within the top-right quadrant.

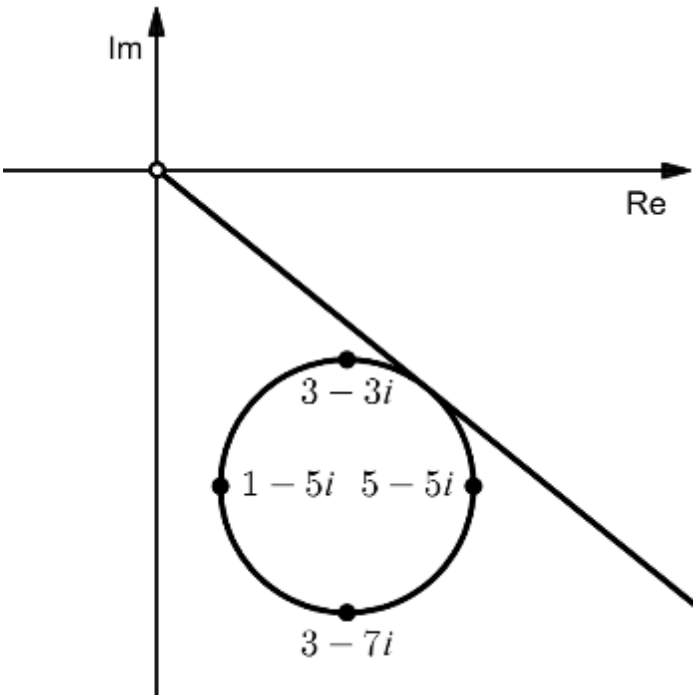


6 b)

$$(x-3)^2 + (y+5)^2 = 4$$

6 c)





Need to find when $y = mx$ and $(x-3)^2 + (y+5)^2 = 4$ have one point of intersection.

$$(x-3)^2 + (mx+5)^2 = 4$$
$$x^2 - 6x + 9 + m^2x^2 + 10mx + 25 - 4 = 0$$
$$(m^2+1)x^2 + (10m-6)x + 30 = 0$$

A single point of intersection occurs when $b^2 - 4ac = 0$

$$(10m-6)^2 - 120(m^2+1) = 0$$
$$100m^2 - 120m + 36 - 120m^2 - 120 = 0$$
$$-20m^2 - 120m - 84 = 0$$
$$\therefore m = \frac{-15 \pm 2\sqrt{30}}{5}$$

So the maximum $\arg z = \arctan\left(\frac{-15+2\sqrt{30}}{5}\right) = -0.680$ (3 SF)

and minimum $\arg z = \arctan\left(\frac{-15-2\sqrt{30}}{5}\right) = -1.38$ (3 SF)

6 d)

The maximum and minimum values of $|z|$ can be found by finding the distance from the origin to the centre, then adding or subtracting the radius:

Maximum $|z| = \sqrt{3^2 + (-5)^2} + 2 = 2 + \sqrt{34}$

Minimum $|z| = \sqrt{3^2 + (-5)^2} - 2 = -2 + \sqrt{34}$

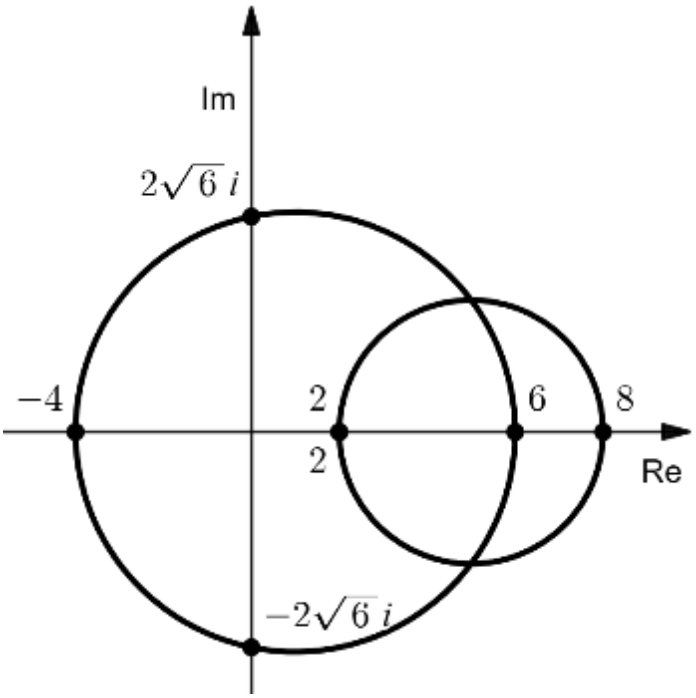
7 a)

Centre is $(1, 0)$ and radius is 5 so $(x-1)^2 + y^2 = 25$

When $x = 0$, $y = \pm 2\sqrt{6}$ and when $y = 0$, $x = -4, 6$

Centre is $(5, 0)$ and radius is 3 so $(x-5)^2 + y^2 = 9$

When $y = 0$, $x = 2, 8$



7 b)

Solve $(x-1)^2 + y^2 = 25$ and $(x-5)^2 + y^2 = 9$ simultaneously:

$(x-1)^2 + y^2 = 25$ gives $y^2 = 25 - (x-1)^2$

Subbing into the other equation:

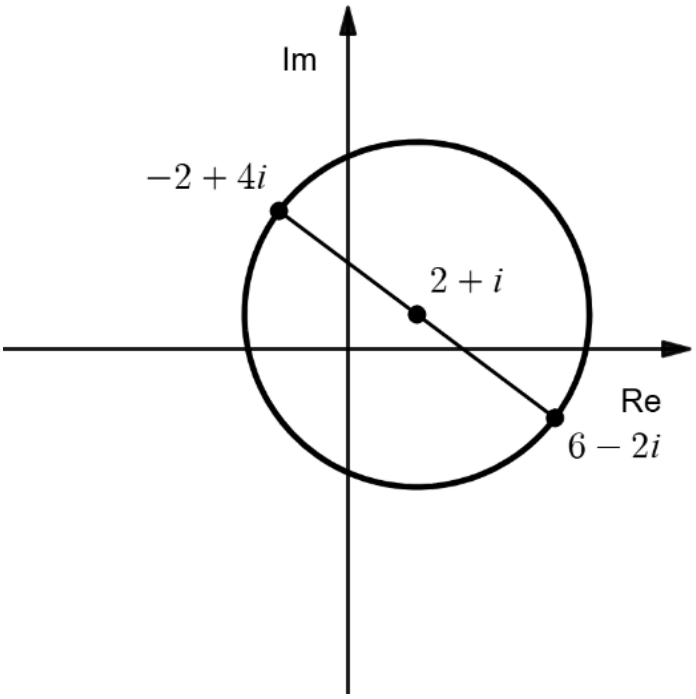
$$(x-5)^2 + 25 - (x-1)^2 = 9$$
$$x^2 - 10x + 25 + 25 - x^2 + 2x - 1 = 9$$
$$8x + 40 = 0 \Rightarrow x = 5$$

So $y = \pm 3$

Therefore, the two values of z are $5 + 3i$ and $5 - 3i$

8 a)

Midpoint is $2 + i$ and radius is $\sqrt{4^2 + 3^2} = 5$



$$|z - 2 - i| = 5$$

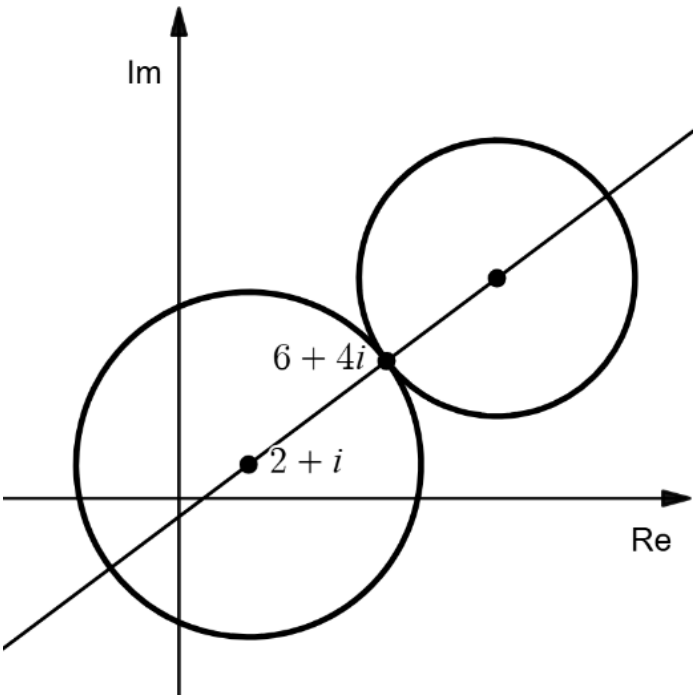
8 b)

$|z - p - qi| = 4$ passes through the point $6 + 4i$

$$|6 - p + 4i - qi| = 2 \Rightarrow \sqrt{(6-p)^2 + (4-q)^2} = 4$$

$$\Rightarrow (6-p)^2 + (4-q)^2 = 16$$

For the loci to meet the first loci once and not cross the axes, it must have a centre on the straight line passing through $2 + i$ and $6 + 4i$



Similar triangles: $\frac{4}{5} \times 4 = \frac{16}{5}$ and $\frac{3}{5} \times 4 = \frac{12}{5}$

$$p = 2 + 4 + \frac{16}{5} = \frac{46}{5}$$

$$q = 1 + 3 + \frac{12}{5} = \frac{32}{5}$$

Alternative Method:

Line passing through $2 + i$ and $6 + 4i$ in cartesian form: $y = \frac{3}{4}x - \frac{1}{2}$

So $q = \frac{3}{4}p - \frac{1}{2}$

$$(6-p)^2 + \left(4 - \left(\frac{3}{4}p - \frac{1}{2}\right)\right)^2 = 16$$

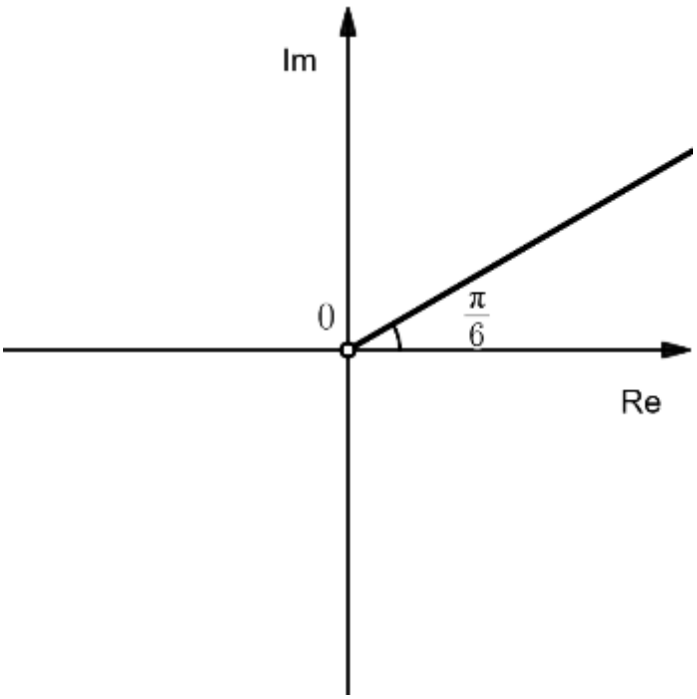
$$(6-p)^2 + \left(\frac{9}{2} - \frac{3}{4}p\right)^2 = 16$$

$$36 - 12p + p^2 + \frac{81}{4} - \frac{27}{4}p + \frac{9}{16}p^2 = 16$$

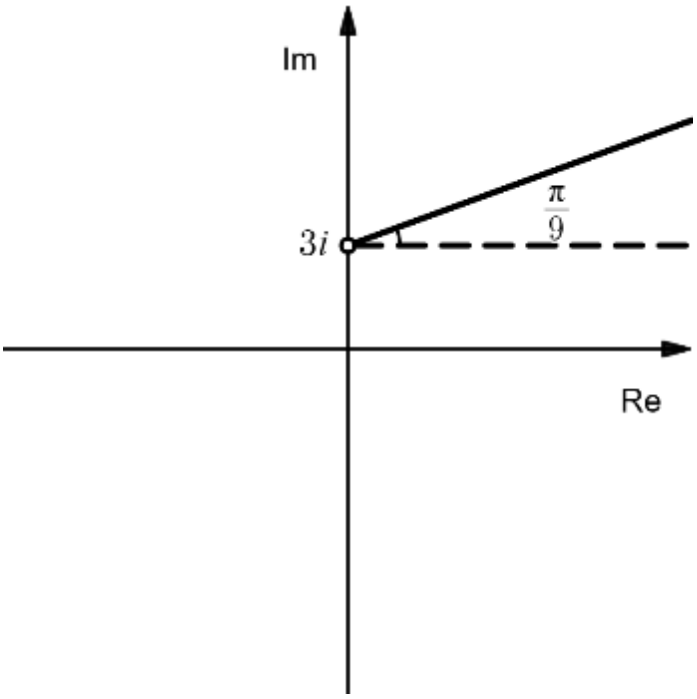
$\frac{25}{16}p^2 - \frac{75}{4}p + \frac{161}{4} = 0$ gives $p = \frac{14}{5}$ or $p = \frac{46}{5}$

so $p = \frac{46}{5}$ and $q = \frac{32}{5}$

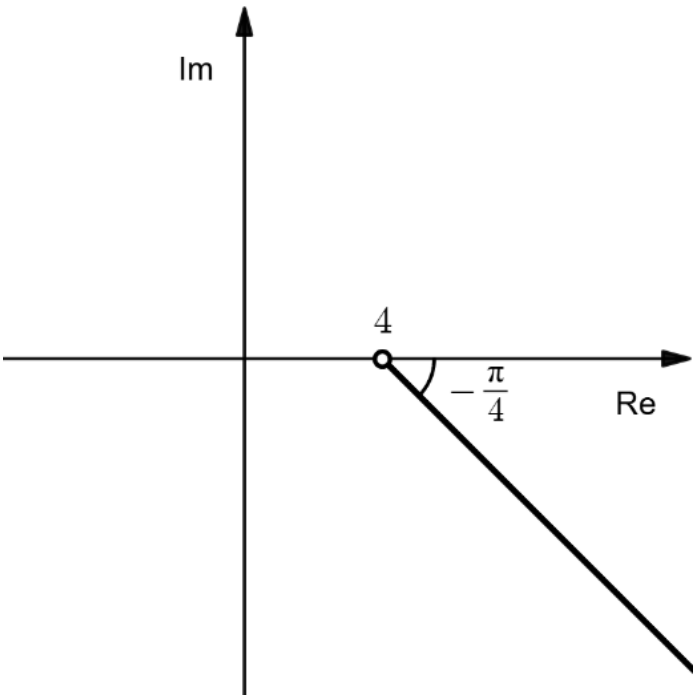
9 a)



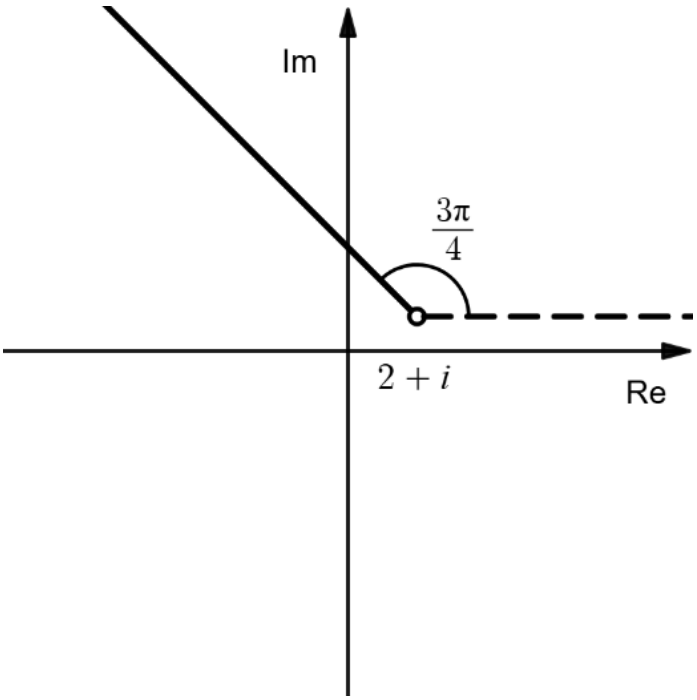
9 b)



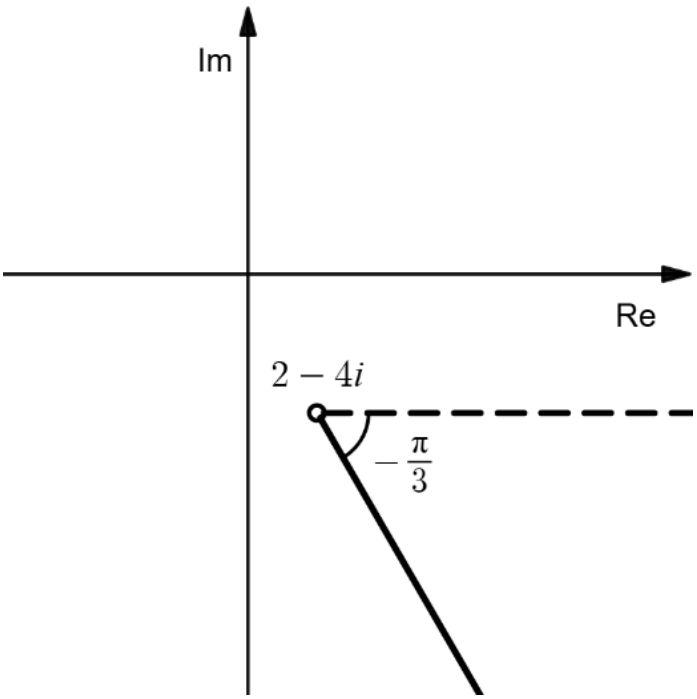
9 c)



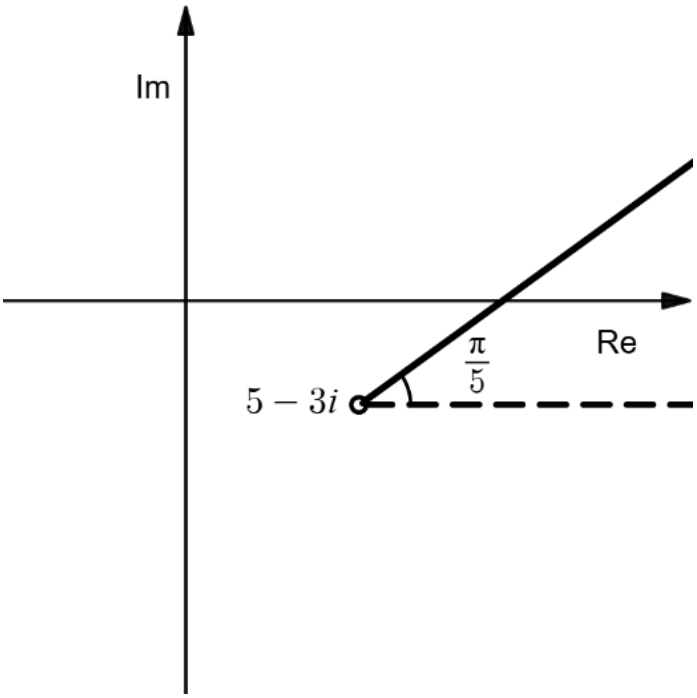
9 d)



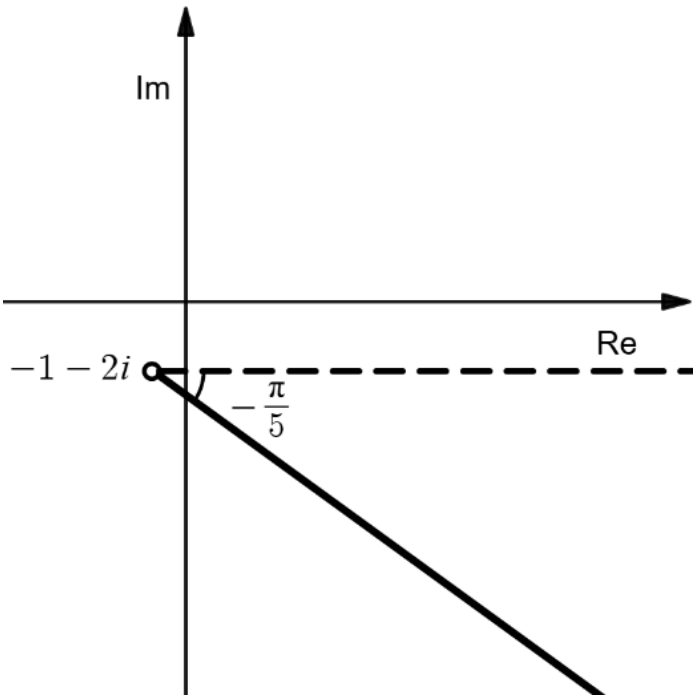
9 e)



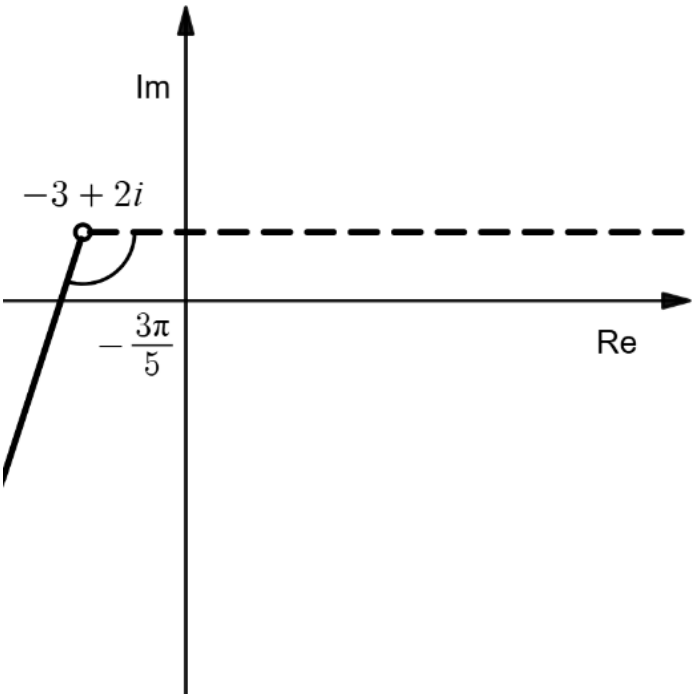
9 f)



9 g)



9 h)



10 a)

$$\arctan(1) = \frac{\pi}{4}$$

$$\arg(z-2) = \frac{\pi}{4}$$

10 b)

$\arctan(1) = \frac{\pi}{4}$ so that $\frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$

$$\arg(z - 3 - i) = \frac{3\pi}{4}$$

10 c)

$$\arctan(1) = \frac{\pi}{4}$$

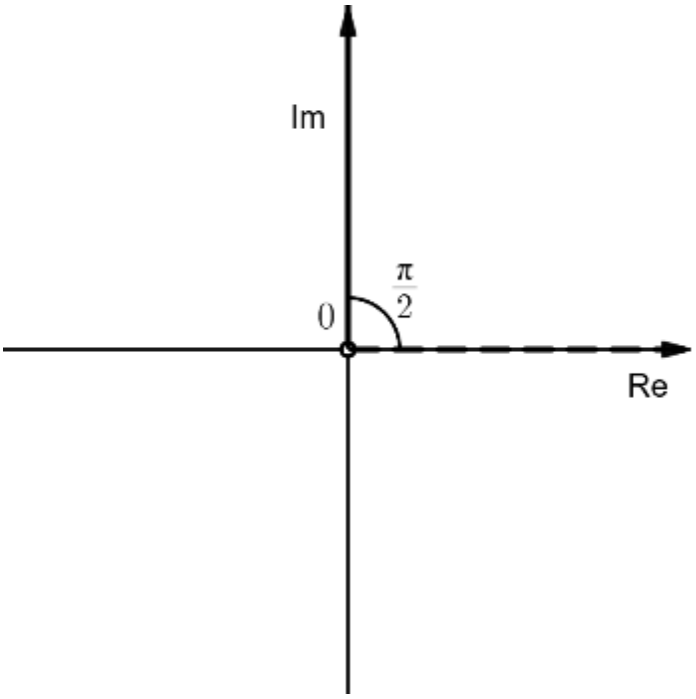
$$\arg(z + 2 - 2i) = -\frac{\pi}{4}$$

10 d)

$\arctan\left(\frac{4}{4\sqrt{3}}\right) = \frac{\pi}{6}$ so that $-\frac{\pi}{2} - \frac{\pi}{6}$

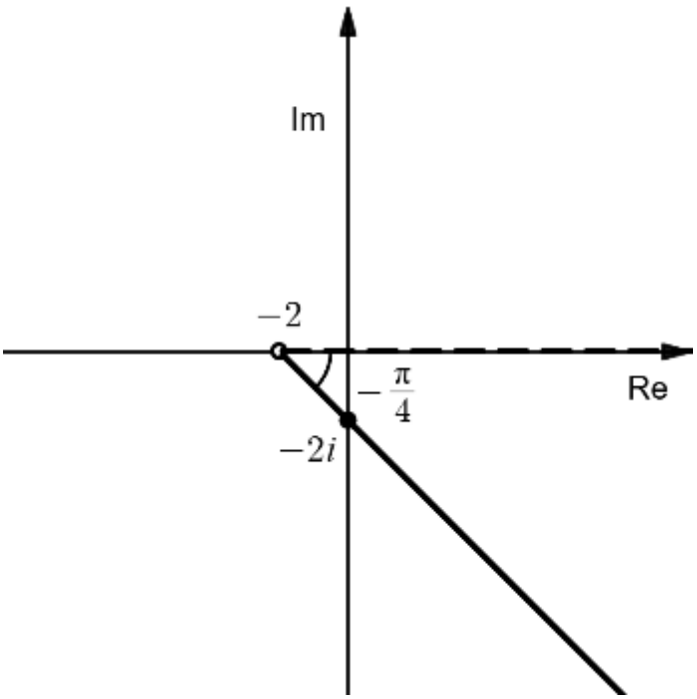
$$\arg(z - 2 - i) = -\frac{2\pi}{3}$$

11 a)



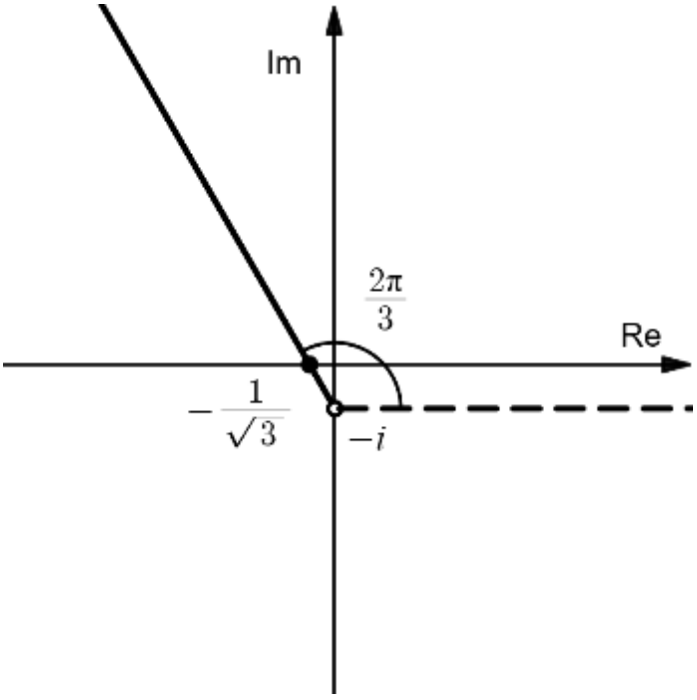
$x = 0$ for $y > 0$

11 b)



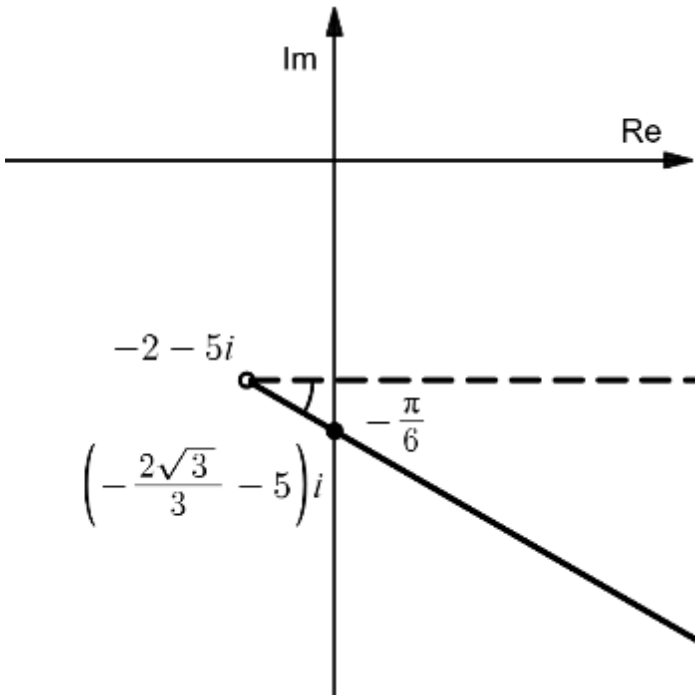
$y + 2 = \tan\left(-\frac{\pi}{4}\right)(x - 0)$ so $y = -x - 2$ for $x > -2$

11 c)



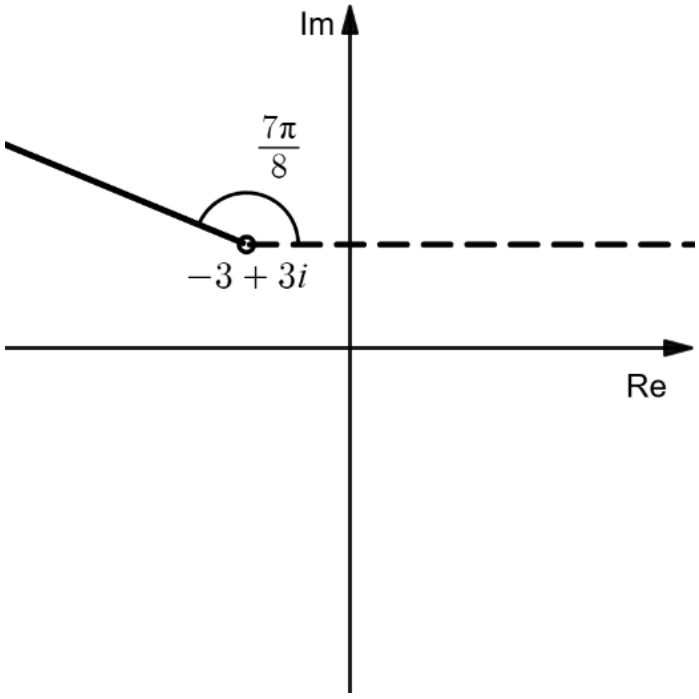
$y + 1 = \tan\left(\frac{2\pi}{3}\right)(x - 0)$ so $y = -\sqrt{3}x - 1$ for $x < 0$

11 d)



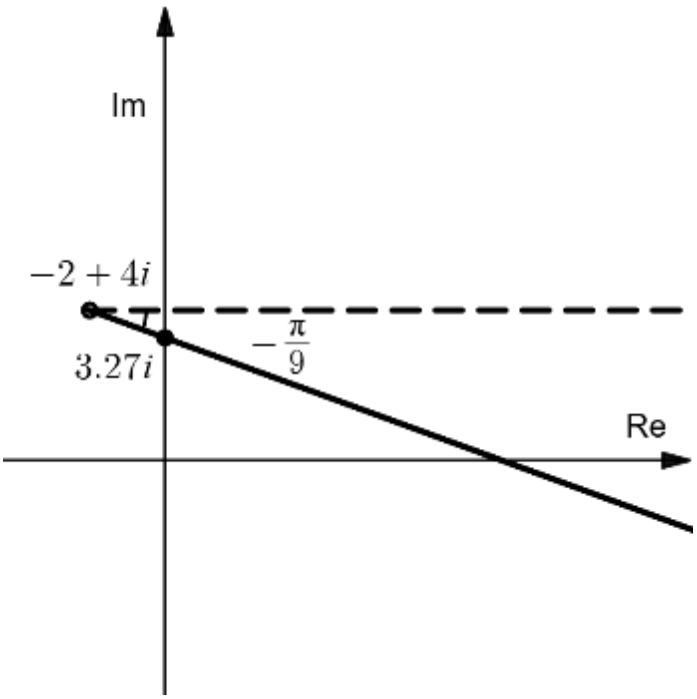
$y+5=\tan\left(-\frac{\pi}{6}\right)(x+2)$ so $y=-\frac{\sqrt{3}}{3}x-\frac{2\sqrt{3}}{3}-5+$ for $x>-2$

11 e)



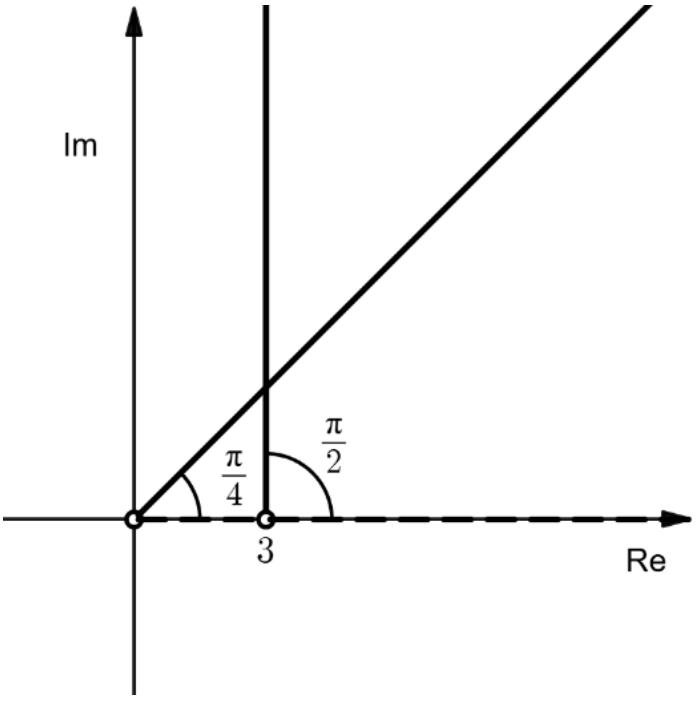
$y-3=\tan\left(\frac{7\pi}{8}\right)(x+3)$ so $y=-0.414x+1.76$ for $x<-3$ (3 SF)

11 f)



$y-4=\tan\left(-\frac{\pi}{9}\right)(x+2)$ so $y=-0.364x+3.27$ for $x>-2$ (3 SF)

12 a)



12 b)

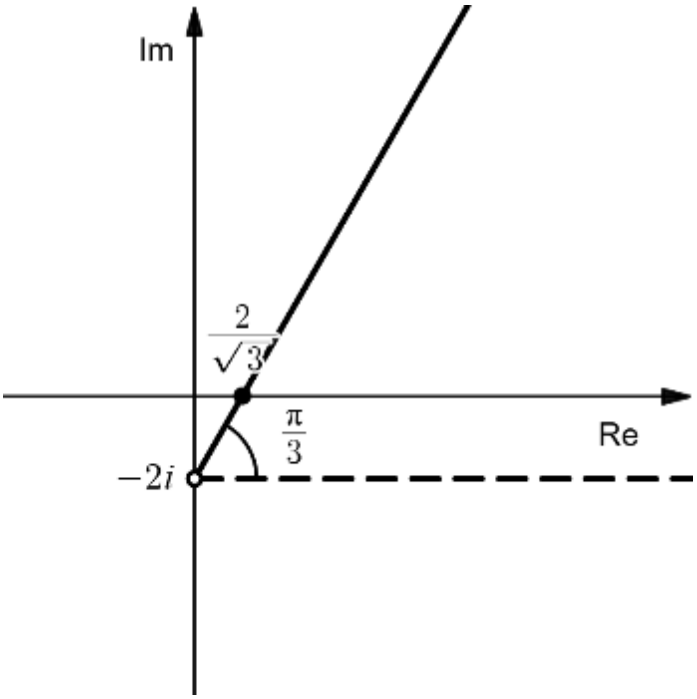
$y=x$ and $x=3$ intersect at $(3,3)$ so $z=3+3i$

12 c)

$$\arg(3+3i-2i)=\arg(3+i)=\arctan\left(\frac{1}{3}\right)=0.322 \text{ (3 SF)}$$

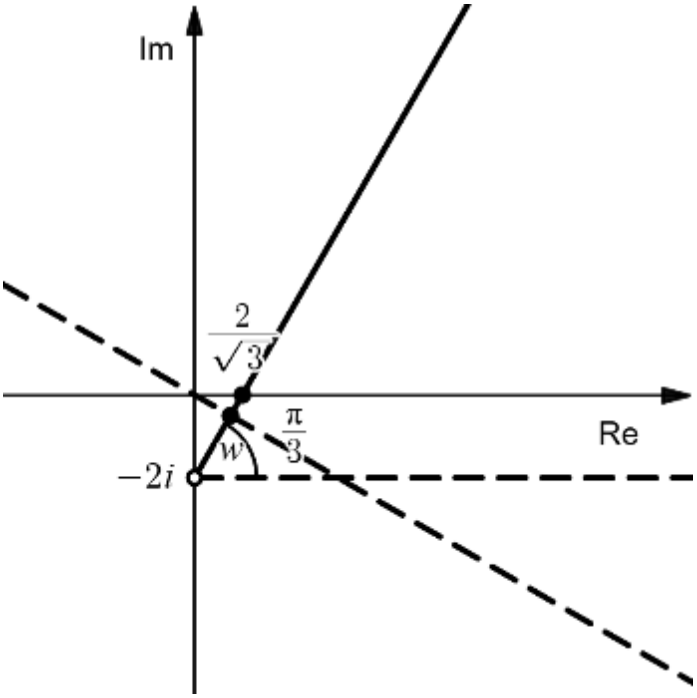
13 a)

$$y + 2 = \tan\left(\frac{\pi}{3}\right)(x - 0) \Rightarrow y = \sqrt{3}x - 2$$



13 b)

$y = \sqrt{3}x - 2$ gives a perpendicular line of $y = -\frac{1}{\sqrt{3}}x$



w is the point that $y = \sqrt{3}x - 2$ and $y = -\frac{1}{\sqrt{3}}x$ intersect:

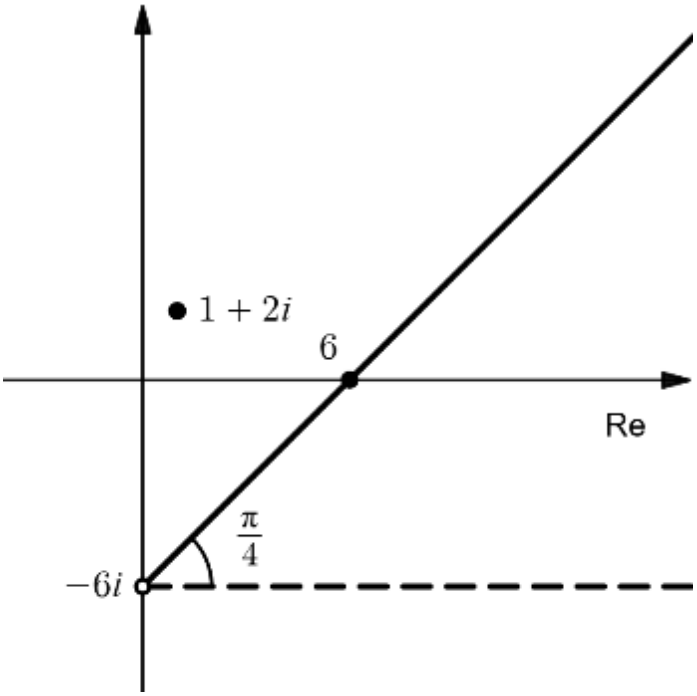
$$\sqrt{3}x - 2 = -\frac{1}{\sqrt{3}}x \Rightarrow x = \frac{\sqrt{3}}{2} \text{ and } y = -\frac{1}{2}$$

Therefore $w = \frac{\sqrt{3}}{2} - \frac{1}{2}i$

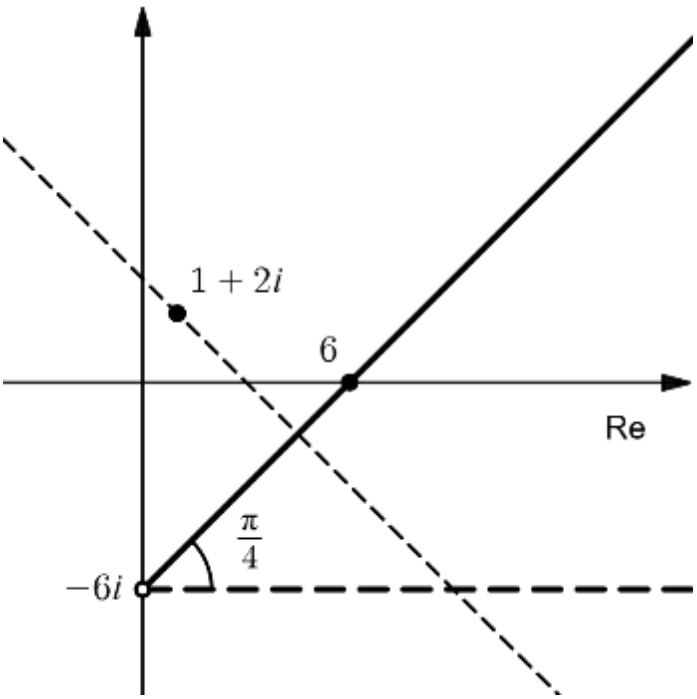
13 c)

$$|z| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = 1$$

14 a)



14 b)



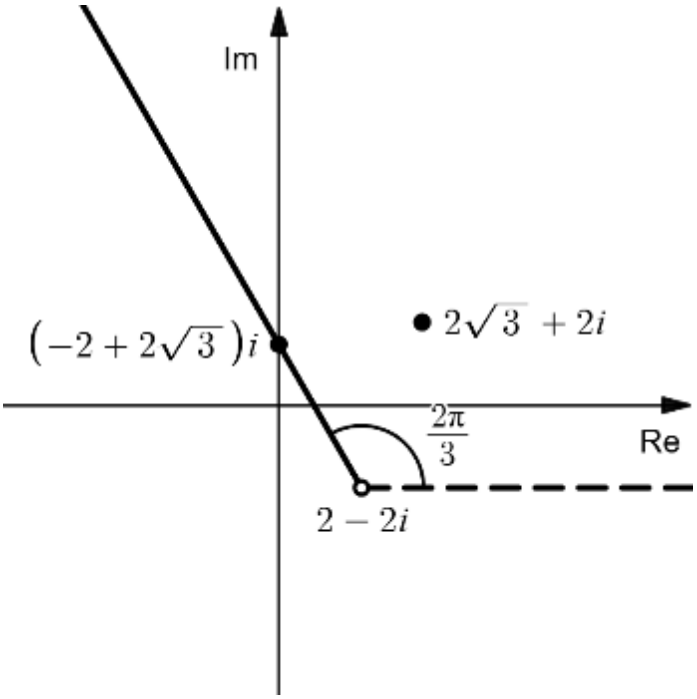
Since $y+6=\tan\left(\frac{\pi}{4}\right)(x-0) \Rightarrow y=x-6$ is the cartesian line of the argument, the perpendicular line which passes through $(1,2)$ is given by

$$y-2=-(x-1) \Rightarrow y=-x+3$$

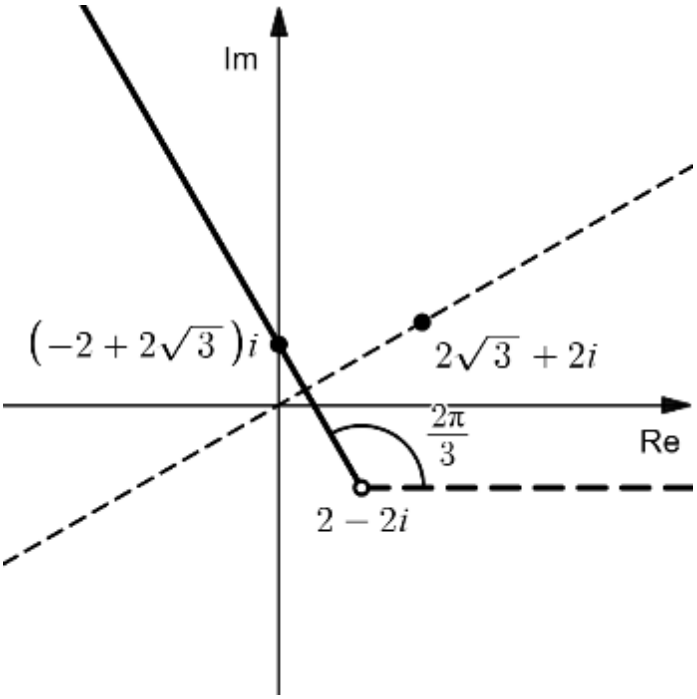
These lines intersect at $\left(\frac{9}{2},-\frac{3}{2}\right)$

Shortest distance between w and $\left(\frac{9}{2},-\frac{3}{2}\right)$ is $\sqrt{\left(\frac{9}{2}-1\right)^2+\left(-\frac{3}{2}-2\right)^2}=\frac{7}{2}\sqrt{2}$

15 a)



15 b)



Since $y+2=\tan\left(\frac{2\pi}{3}\right)(x-2)\Rightarrow y=-\sqrt{3}x+2\sqrt{3}-2$ is the cartesian line of the argument, the perpendicular line which passes through $(2\sqrt{3},2)$ is given by

$$y-2=\frac{1}{\sqrt{3}}(x-2\sqrt{3})\Rightarrow y=\frac{1}{\sqrt{3}}x$$

These lines intersect at $\left(\frac{3}{2}-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$

Shortest distance between w and $\left(\frac{3}{2}-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$ is

$$\begin{aligned}&\sqrt{\left(2\sqrt{3}-\frac{3}{2}+\frac{\sqrt{3}}{2}\right)^2+\left(2-\frac{\sqrt{3}}{2}+\frac{1}{2}\right)^2}\\&=\sqrt{\left(\frac{5}{2}\sqrt{3}-\frac{3}{2}\right)^2+\left(\frac{5}{2}-\frac{\sqrt{3}}{2}\right)^2}\\&=\sqrt{21-\frac{15}{2}\sqrt{3}+7-\frac{5}{2}\sqrt{3}}=\sqrt{28-10\sqrt{3}}\end{aligned}$$

$$\sqrt{28-10\sqrt{3}}=a+b\sqrt{3}\Rightarrow 28-10\sqrt{3}=(a+b\sqrt{3})^2=a^2+2ab\sqrt{3}+3b^2$$

$2ab=-10$ and $a^2+3b^2=28$

Solving simultaneously gives

$$b=-\frac{5}{a}\Rightarrow a^2+3\left(-\frac{5}{a}\right)^2=28\Rightarrow a^4+75=28a^2\Rightarrow a^4-28a^2+75=0$$

$\therefore a^2=25$ or $a^2=3$ but a is an integer so $a=\pm5$

So $a=\pm5$, $b=\mp1$

Since it is a distance $a=5$, $b=-1$ so $5-\sqrt{3}$

16 a)

Points: 4 and i gives $|z-4|=|z-i|$

Midpoint: $\left(2,\frac{1}{2}\right)$ and gradient between points is $-\frac{1}{4}$

$$\therefore y-\frac{1}{2}=4(x-2)\Rightarrow y=4x-\frac{15}{2}$$

16 b)

Points: 2 and $-3i$ gives $|z-2|=|z+3i|$

Midpoint: $\left(1,-\frac{3}{2}\right)$ and gradient between points is $\frac{3}{2}$

$$\therefore y+\frac{3}{2}=-\frac{2}{3}(x-1)\Rightarrow y=-\frac{2}{3}x-\frac{5}{6}$$

16 c)

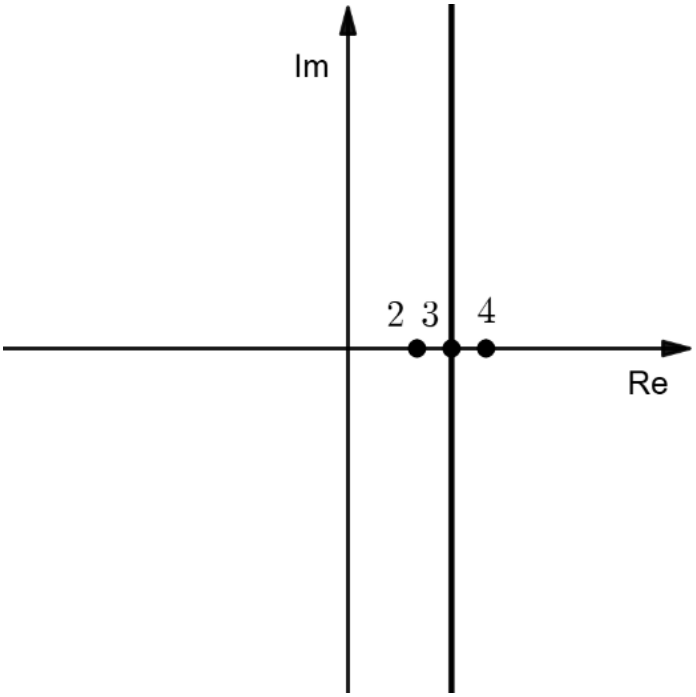
Points: $4-2i$ and $2-4i$ gives $|z-4+2i|=|z-2+4i|$

Midpoint: $(3,-3)$ and gradient between points is 1

$$\therefore y+3=-1(x-3)\Rightarrow y=-x$$

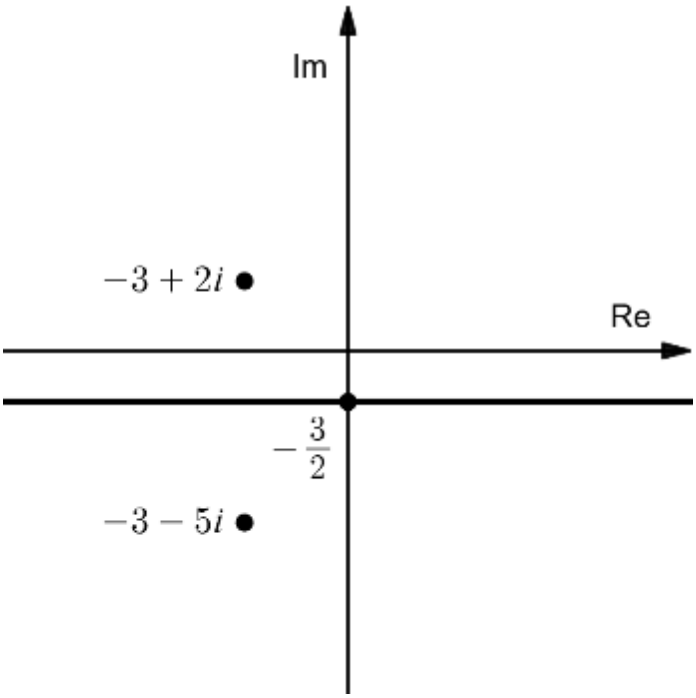
17 a)

$x = 3$



17 b)

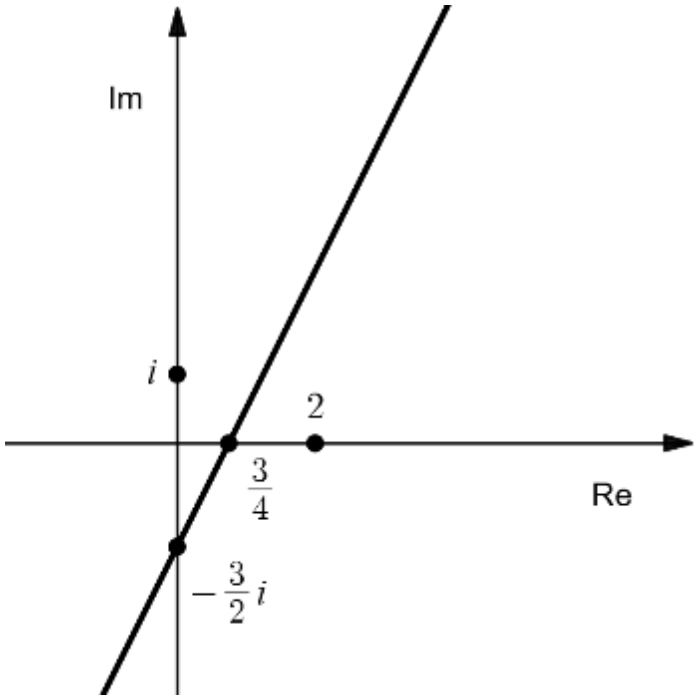
$y = -\frac{3}{2}$



17 c)

Points: 2 and i gives midpoint: $\left(1, \frac{1}{2}\right)$ and gradient between points is $-\frac{1}{2}$

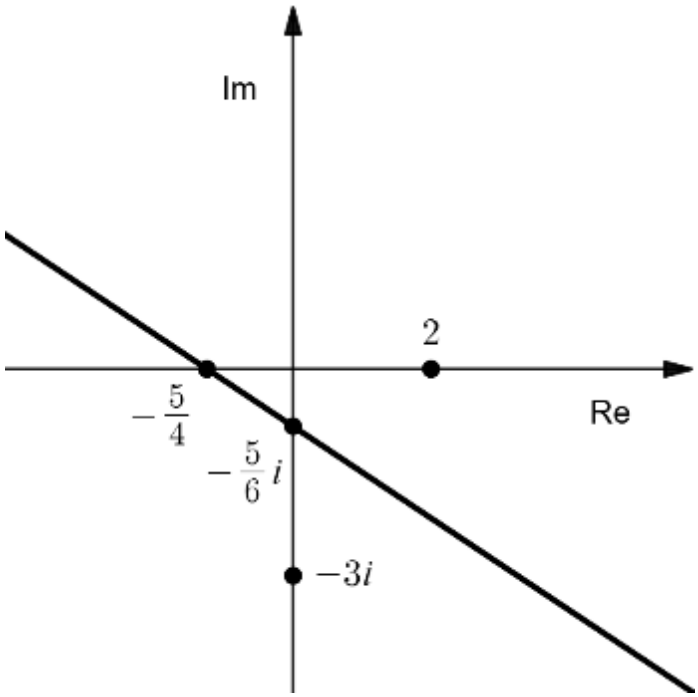
$$\therefore y - \frac{1}{2} = 2(x-1) \Rightarrow y = 2x - \frac{3}{2}$$



17 d)

Points: 2 and $-3i$ gives midpoint: $\left(1, -\frac{3}{2}\right)$ and gradient between points is $\frac{3}{2}$

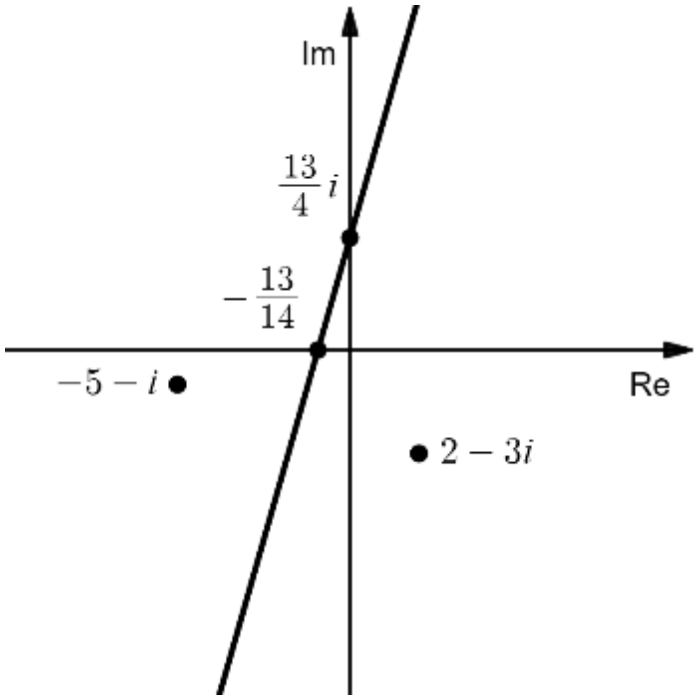
$$\therefore y + \frac{3}{2} = -\frac{2}{3}(x-1) \Rightarrow y = -\frac{2}{3}x - \frac{5}{6}$$



17 e)

Points: $-5-i$ and $2-3i$ gives midpoint: $\left(-\frac{3}{2},-2\right)$ and gradient between points is $-\frac{2}{7}$

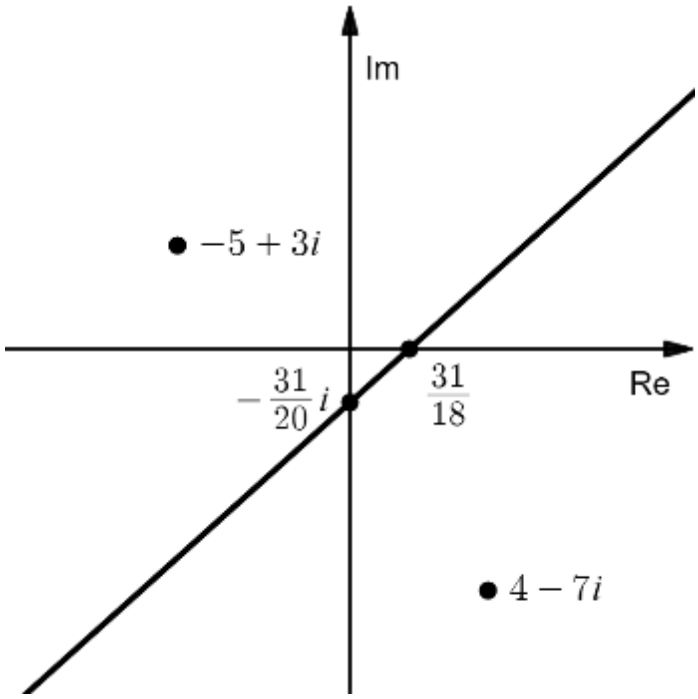
$$\therefore y+2=\frac{7}{2}\left(x+\frac{3}{2}\right)\Rightarrow y=\frac{7}{2}x+\frac{13}{4}$$



17 f)

Points: $-5+3i$ and $4-7i$ gives midpoint: $\left(-\frac{1}{2},-2\right)$ and gradient between points is $-\frac{10}{9}$

$$\therefore y+2=\frac{9}{10}\left(x+\frac{1}{2}\right)\Rightarrow y=\frac{9}{10}x-\frac{31}{20}$$



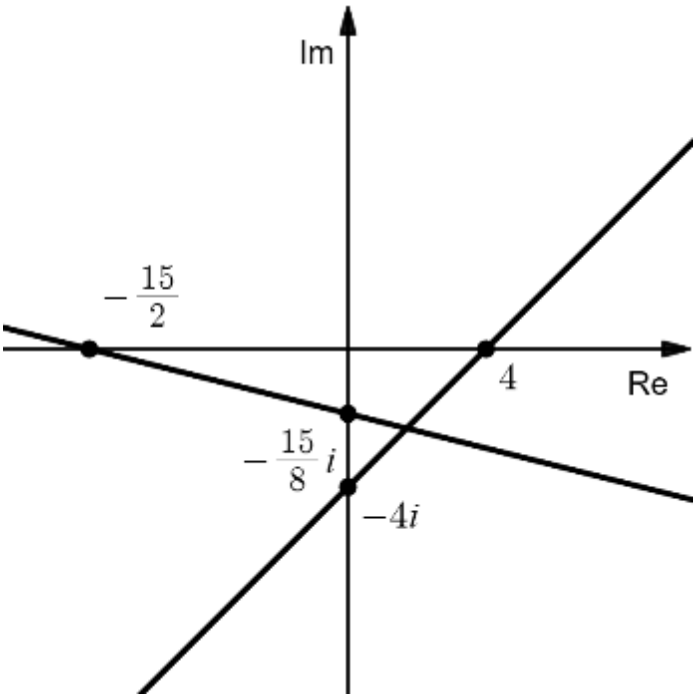
18 a)

Points: $2-i$ and $3-2i$ gives midpoint: $\left(\frac{5}{2},-\frac{3}{2}\right)$ and gradient between points is -1

$$\therefore y+\frac{3}{2}=1\left(x-\frac{5}{2}\right)\Rightarrow y=x-4$$

Points: 1 and $-4i$ gives midpoint: $\left(\frac{1}{2},-2\right)$ and gradient between points is 4

$$\therefore y+2=-\frac{1}{4}\left(x-\frac{1}{2}\right)\Rightarrow y=-\frac{1}{4}x-\frac{15}{8}$$



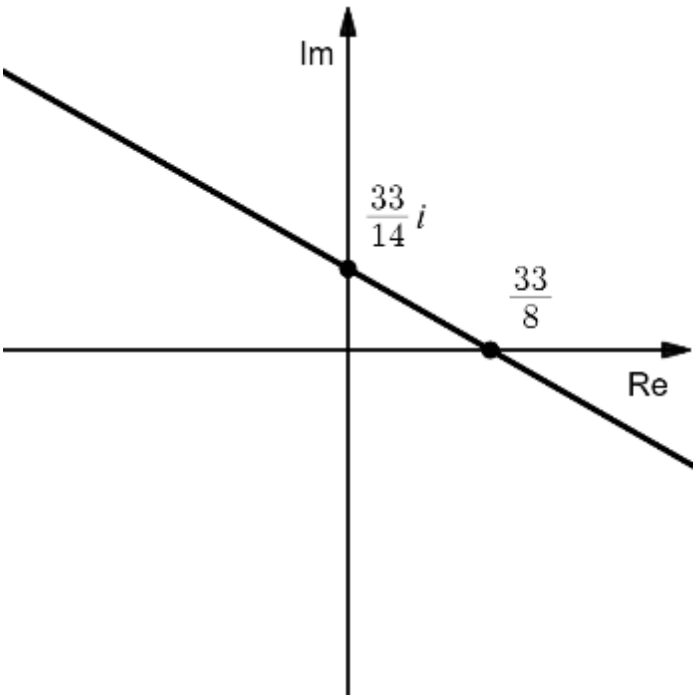
18 b)

$y=x-4$ and $y=-\frac{1}{4}x-\frac{15}{8}$ gives $x-4=-\frac{1}{4}x-\frac{15}{8}$ gives $x=\frac{17}{10}$ and $y=-\frac{23}{10}$
so $\frac{17}{10}-\frac{23}{10}i$

19 a)

Points: -4 and $7i$ gives midpoint: $\left(-2,\frac{7}{2}\right)$ and gradient between points is $\frac{7}{4}$

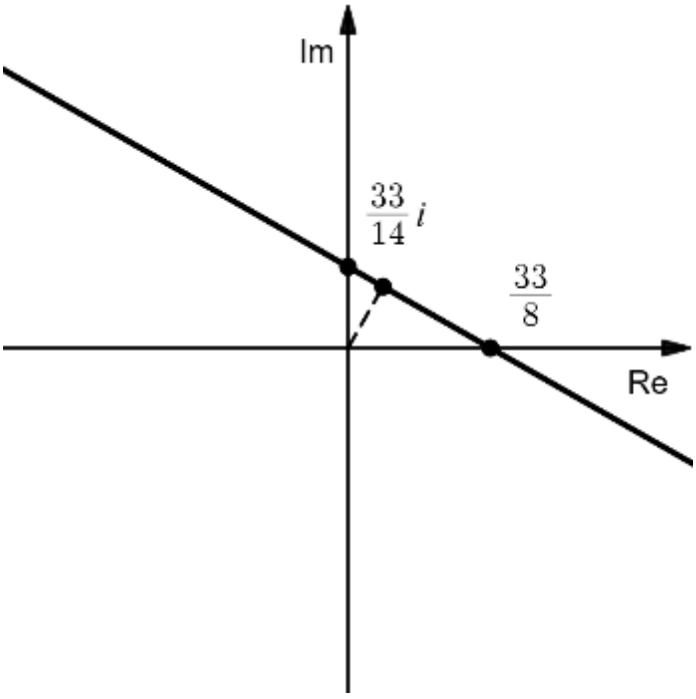
$$\therefore y-\frac{7}{2}=-\frac{4}{7}(x+2)\Rightarrow y=-\frac{4}{7}x+\frac{33}{14}$$



19 b)

$$y = -\frac{4}{7}x + \frac{33}{14}$$

19 c)



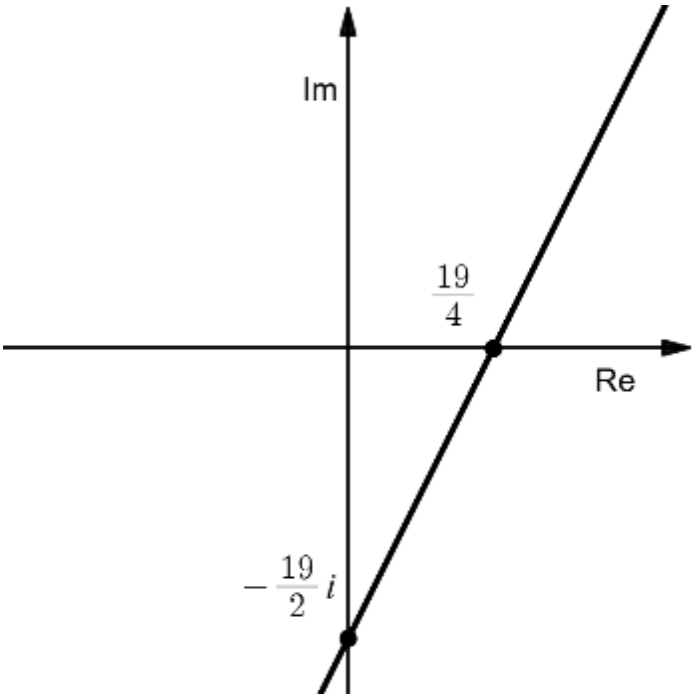
$y = -\frac{4}{7}x + \frac{33}{14}$ and $y = \frac{7}{4}x$ intersect when $-\frac{4}{7}x + \frac{33}{14} = \frac{7}{4}x$ which gives $\left(\frac{66}{65}, \frac{231}{130}\right)$

$$\sqrt{\left(\frac{66}{65}\right)^2 + \left(\frac{231}{130}\right)^2} = \frac{33}{130}\sqrt{65}$$

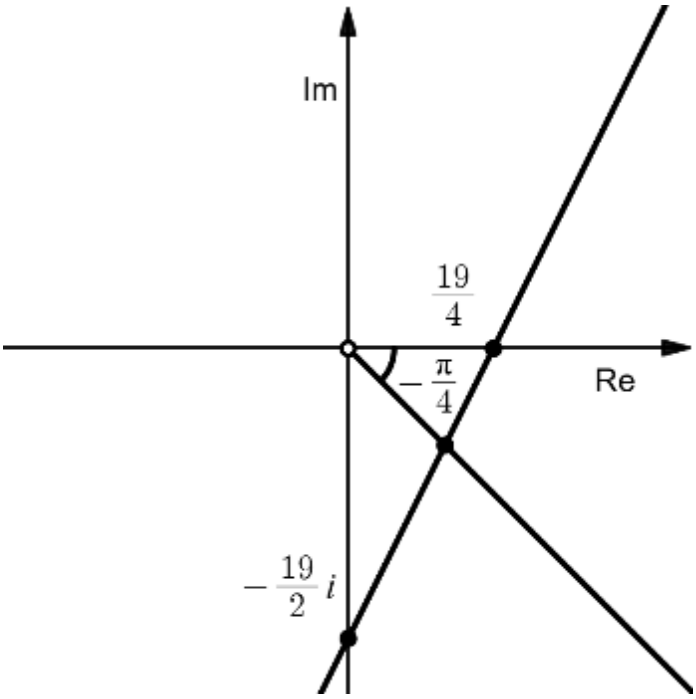
20 a)

Points: $2-3i$ and $4-4i$ gives midpoint: $\left(3,-\frac{7}{2}\right)$ and gradient between points is $-\frac{1}{2}$

$$\therefore y+\frac{7}{2}=2(x-3)\Rightarrow y=2x-\frac{19}{2}$$



20 b)



$\arg(z)=\frac{\pi}{4}$ has cartesian equation $y=-x$

$-x=2x-\frac{19}{2}$ gives $x=\frac{19}{6}$

So z is $\frac{19}{6}-\frac{19}{6}i$

21 a)

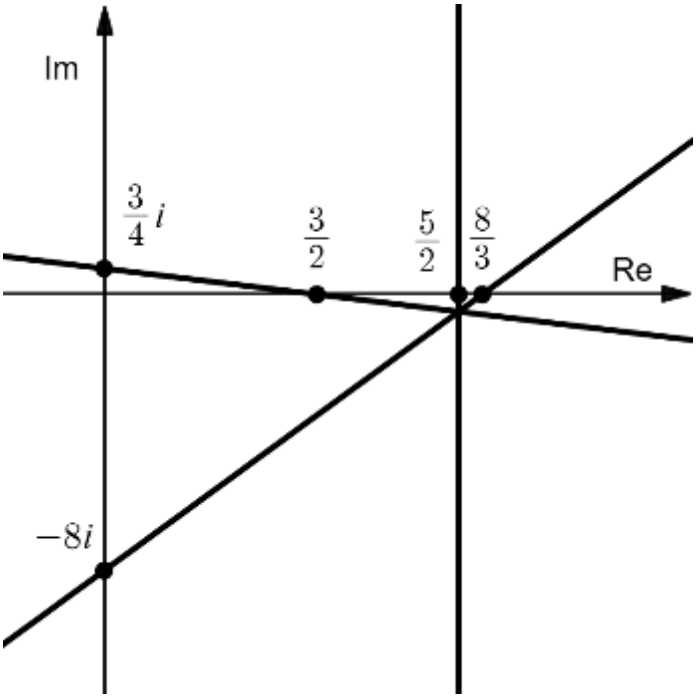
Points: -1 and 6 gives midpoint: $\left(0, \frac{5}{2}\right)$ so $x = \frac{5}{2}$

Points: $2i$ and 6 gives midpoint: $(3, 1)$ and gradient between points is $-\frac{1}{3}$

$$\therefore y - 1 = 3(x - 3) \Rightarrow y = 3x - 8$$

Points: -1 and $2i$ gives midpoint: $\left(-\frac{1}{2}, 1\right)$ and gradient between points is 2

$$\therefore y - 1 = -\frac{1}{2}\left(x + \frac{1}{2}\right) \Rightarrow y = -\frac{1}{2}x + \frac{3}{4}$$



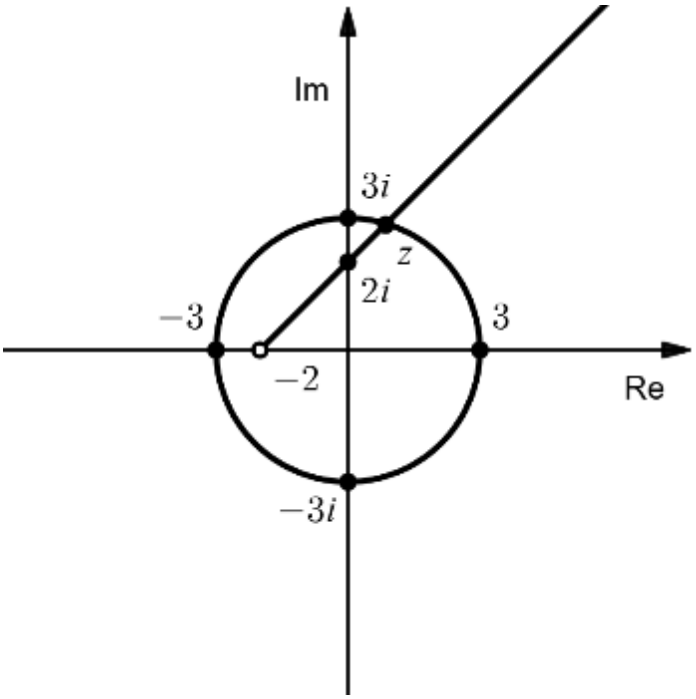
21 b)

$x = \frac{5}{2}$ and $y = 3x - 8$ and $y = -\frac{1}{2}x + \frac{3}{4}$

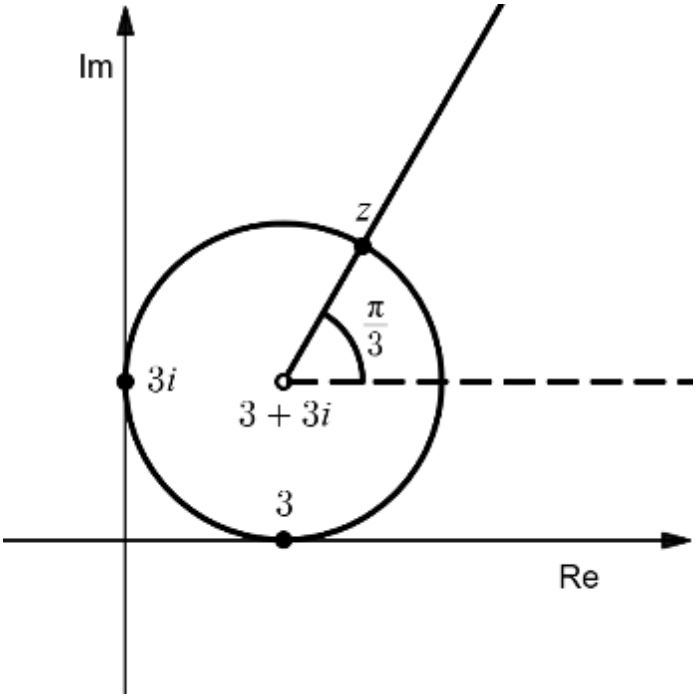
$x = \frac{5}{2}$ gives $y = 3 \times \frac{5}{2} - 8 = -\frac{1}{2}$

So $z = \frac{5}{2} - \frac{1}{2}i$

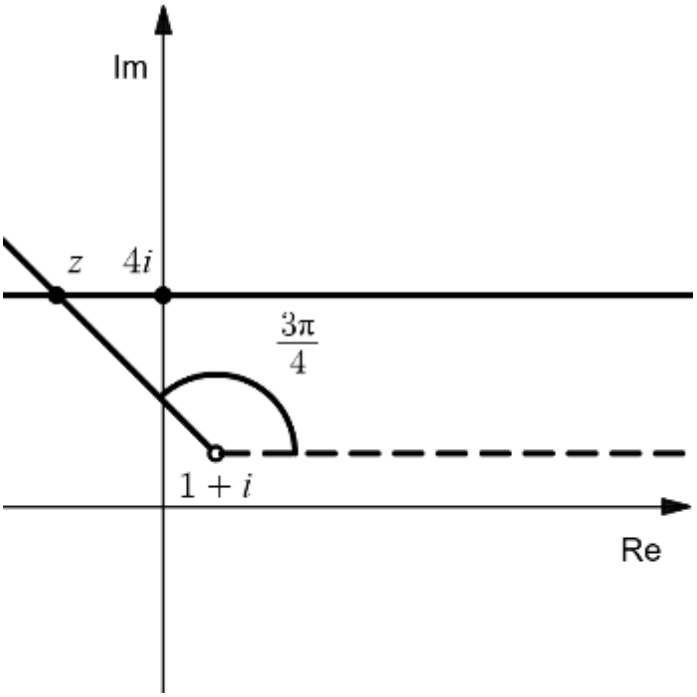
22 a)



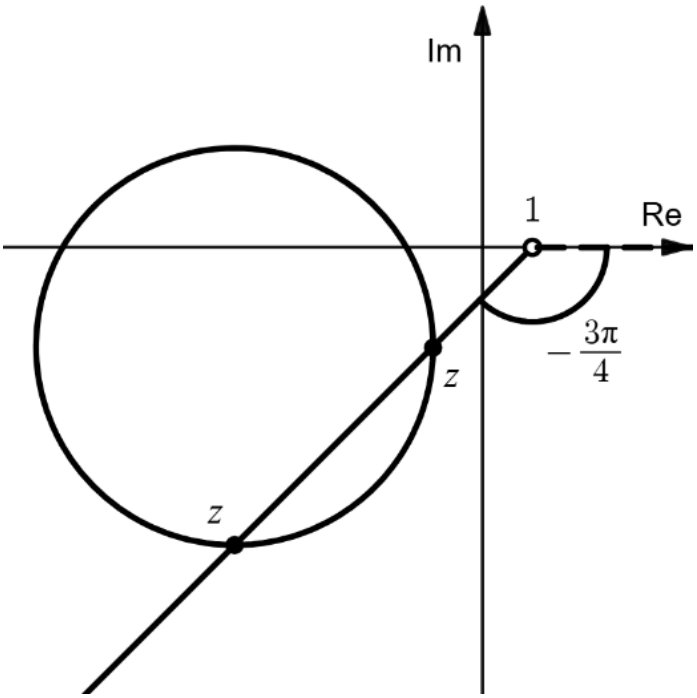
22 b)



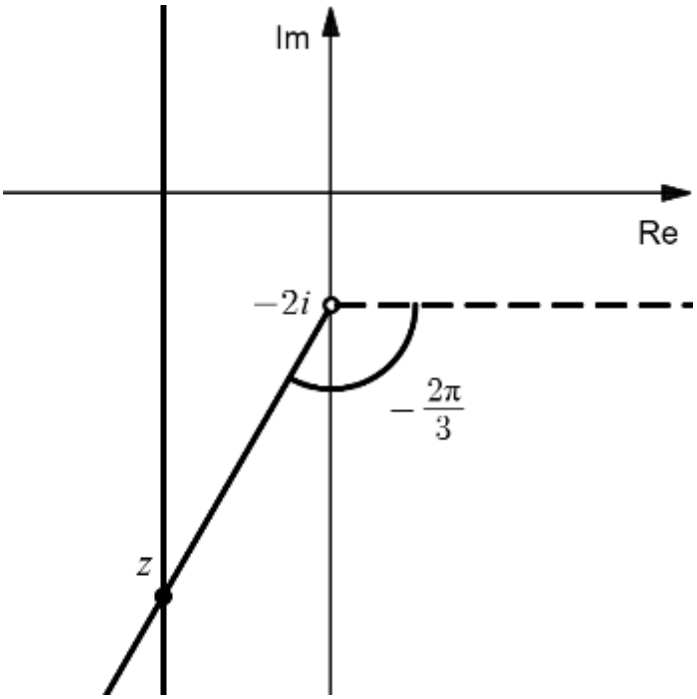
22 c)



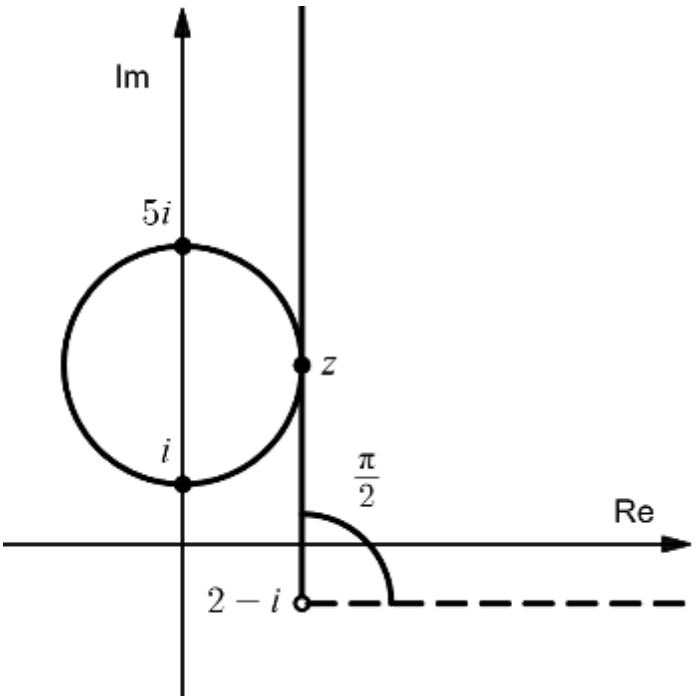
22 d)



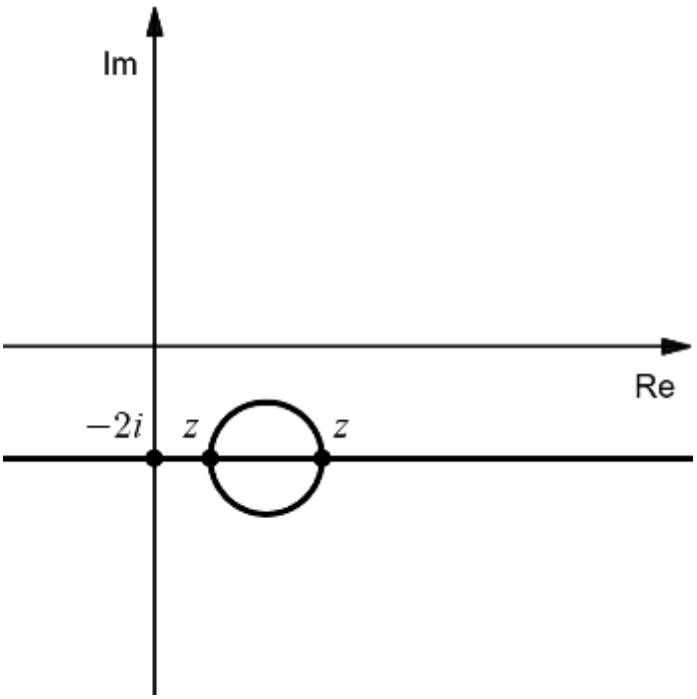
22 e)



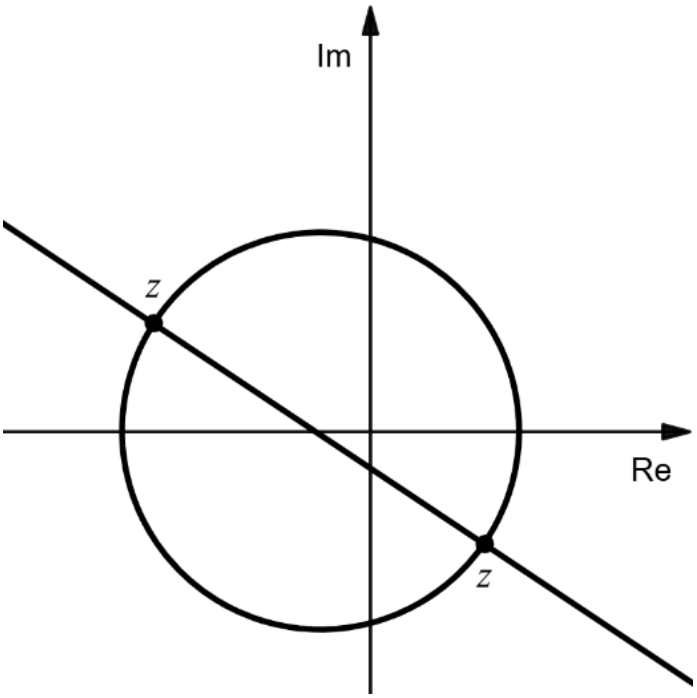
22 f)



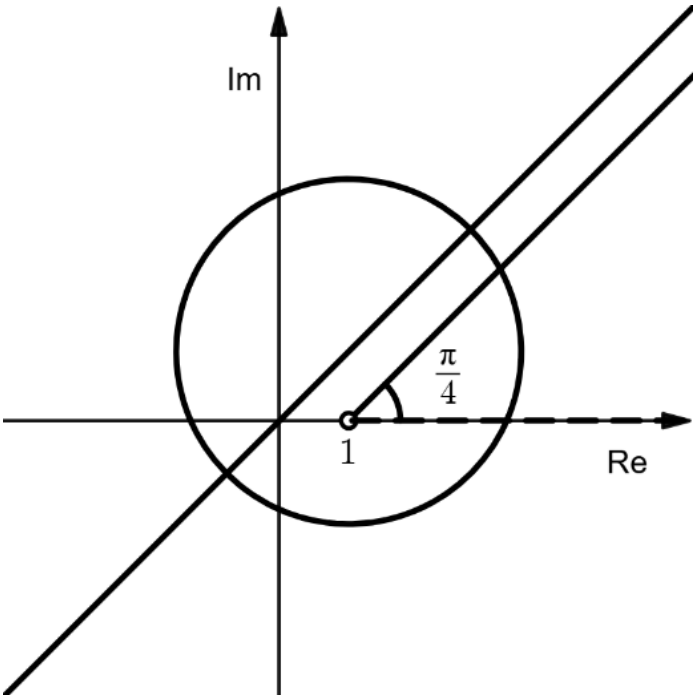
22 g)



22 h)

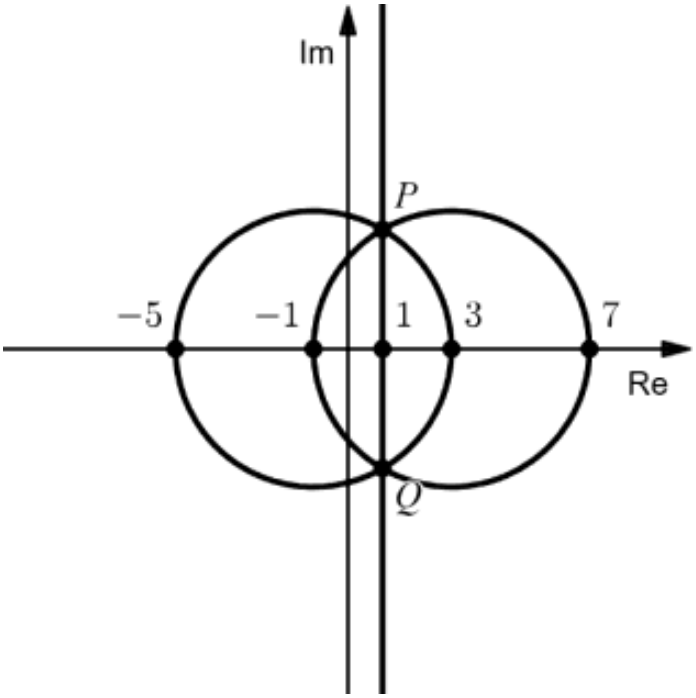


22 i)



There are no values of z for which all the conditions hold.

23)



24 a)

Centre at 1, radius of 5 so $|z-1|=5$

24 b)

Angle is $\arctan(1)=\frac{\pi}{4}$ so $\arg(z+2)=\frac{\pi}{4}$

24 c)

Solve $(x-1)^2+y^2=5^2$ and $y=x+2$ simultaneously:

$$(x-1)^2+(x+2)^2=5^2$$
$$x^2-2x+1+x^2+4x+4=25$$

$2x^2+2x-20=0$ which gives $x=-\frac{1}{2}\pm\frac{\sqrt{41}}{2}$

Choosing the positive solution:

$x=-\frac{1}{2}+\frac{\sqrt{41}}{2}$ and $y=\frac{3}{2}\pm\frac{\sqrt{41}}{2}$

$$\therefore z=-\frac{1}{2}+\frac{\sqrt{41}}{2}+\left(\frac{3}{2}+\frac{\sqrt{41}}{2}\right)i$$

25 a)

Centre at $1-3i$, radius of 3 so $|z-1+3i|=3$

25 b)

$$|z+4-3i|=|z+2-4i|$$

25 c)

$y-\frac{7}{2}=-2(x+3)\Rightarrow y=-2x-\frac{5}{2}$ and $(x-1)^2+(y+3)^2=9$

Solve $y=-2x-\frac{5}{2}$ and $(x-1)^2+(y+3)^2=9$ simultaneously:

$$(x-1)^2+\left(-2x-\frac{5}{2}+3\right)^2=9$$
$$(x-1)^2+\left(-2x+\frac{1}{2}\right)^2=9$$
$$x^2-2x+1+4x^2-2x+\frac{1}{4}=9$$

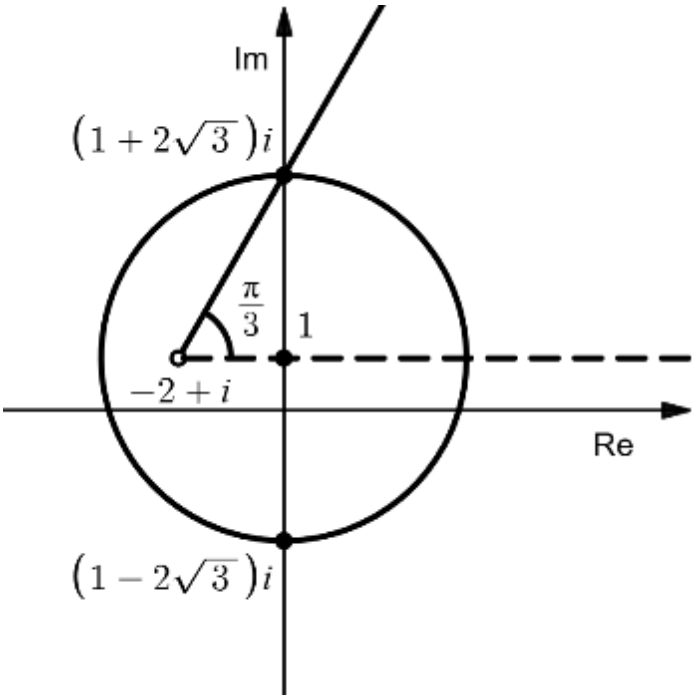
$5x^2-4x-\frac{31}{4}=0$ which gives $x=\frac{2}{5}\pm\frac{3}{10}\sqrt{19}$ and $y=-\frac{33}{10}\mp\frac{3}{5}\sqrt{19}$

$\frac{2}{5}+\frac{3}{10}\sqrt{19}+\left(-\frac{33}{10}-\frac{3}{5}\sqrt{19}\right)i$ and $\frac{2}{5}-\frac{3}{10}\sqrt{19}+\left(-\frac{33}{10}+\frac{3}{5}\sqrt{19}\right)i$

26)

Circles crosses imaginary axis at $(1+2\sqrt{3})i$ and $(1-2\sqrt{3})i$

$2\tan\left(\frac{\pi}{3}\right)=2\sqrt{3}$ so the line passes through at $1+2\sqrt{3}$



$z = (1 + 2\sqrt{3})i$

27)

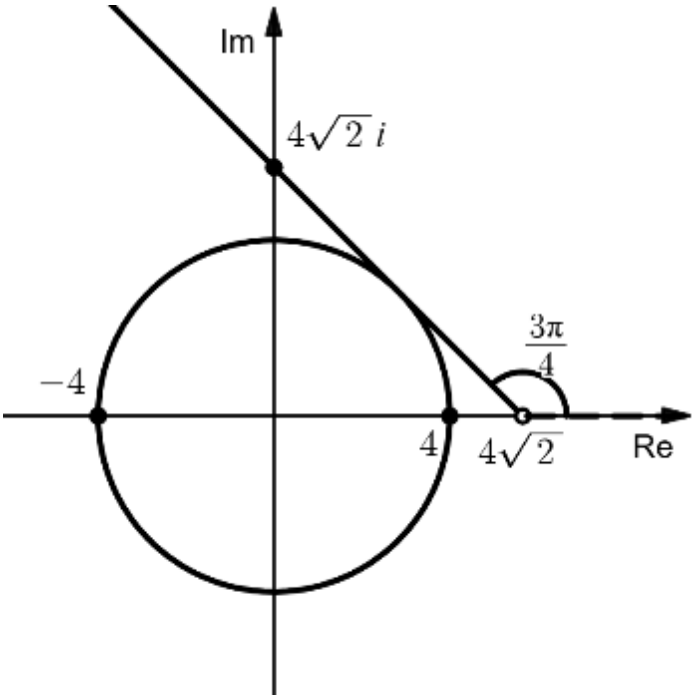
Circle is centred at origin, with radius $4 \Rightarrow x^2 + y^2 = 16$

Cartesian equation of the half line: $y - 0 = -(x - 4\sqrt{2}) \Rightarrow y = -x + 4\sqrt{2}$

$$x^2 + (-x + 4\sqrt{2})^2 = 16$$

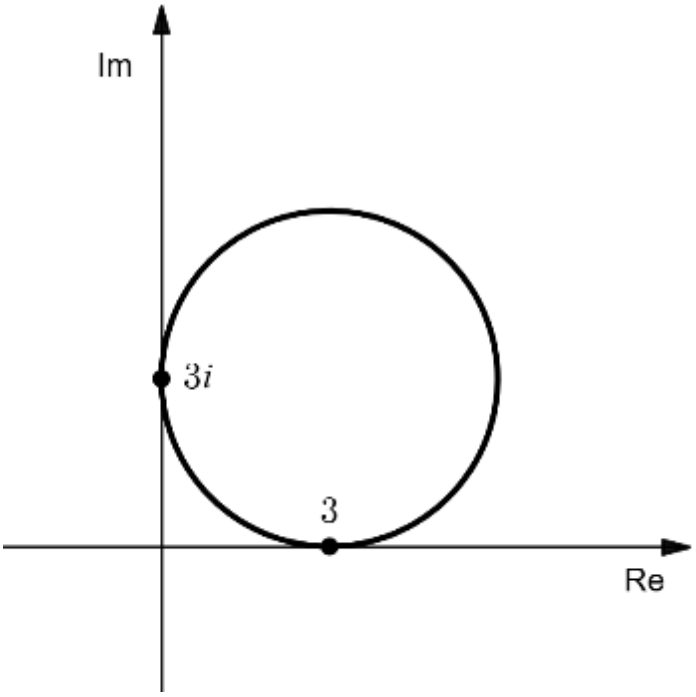
$$x^2 + x^2 - 8\sqrt{2}x + 32 = 16$$

$2x^2 - 8\sqrt{2}x + 16 = 0$ which gives $x = 2\sqrt{2}$ and $y = 2\sqrt{2}$



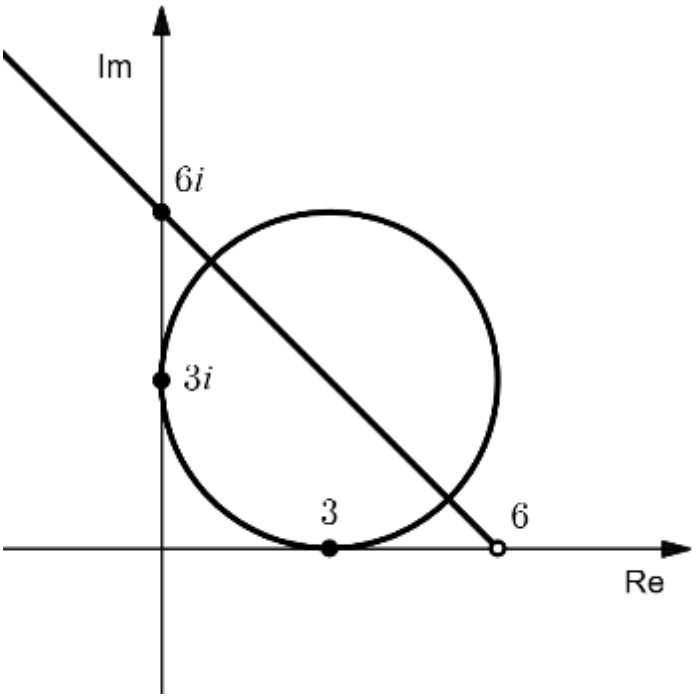
$z = 2\sqrt{2} + 2\sqrt{2}i$

28 a)



28 b)

$\arg(z-6) = \frac{3\pi}{4}$ is given by $y = -x + 6$ for $x < 3$



Solve $y = -x + 6$ and $(x-3)^2 + (y-3)^2 = 9$ simultaneously:

$$(x-3)^2 + (-x+6-3)^2 = 9$$

$$(x-3)^2 + (-x+3)^2 = 9$$

$$x^2 - 6x + 9 + x^2 - 6x + 9 = 9$$

$$2x^2 - 12x + 9 = 0 \Rightarrow x = 3 \pm \frac{3\sqrt{2}}{2} \text{ and } y = 3 \mp \frac{3\sqrt{2}}{2}$$

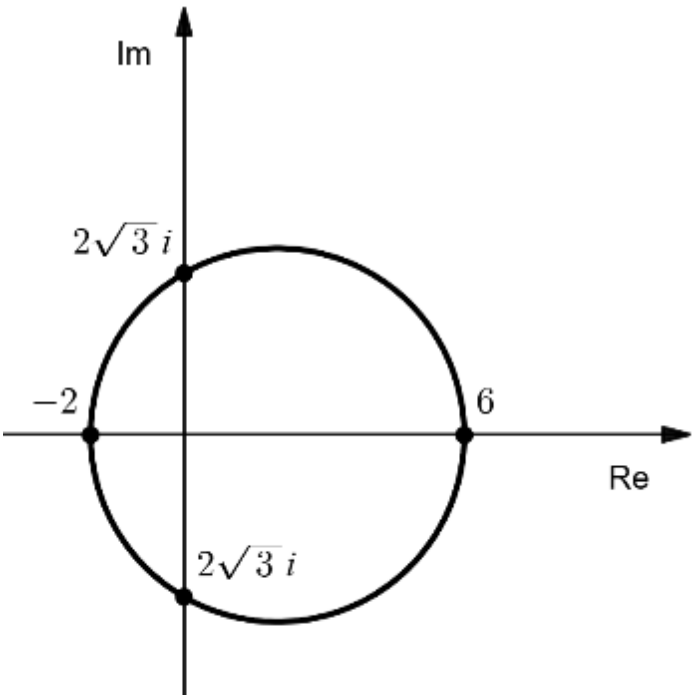
$$\sqrt{\left(3+\frac{3\sqrt{2}}{2}\right)^2+\left(3-\frac{3\sqrt{2}}{2}\right)^2}=3\sqrt{3}$$

$$|z|=3\sqrt{3}$$

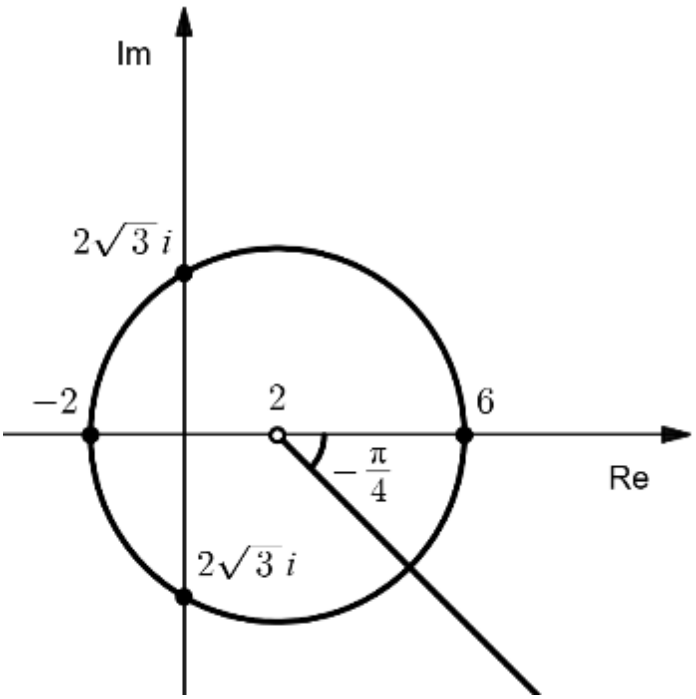
29 a)

$$(x-2)^2+y^2=16$$

When $x=0$, $y=\sqrt{12}=2\sqrt{2}$



29 b)



Solve $(x-2)^2+y^2=16$ and $y=-x+2$ for $x>2$ simultaneously:

$$(x-2)^2+(-x+2)^2=16$$
$$x^2-4x+4+x^2-4x+4=16$$

$2x^2-8x-8=0$ gives $x=2\pm 2\sqrt{2}$

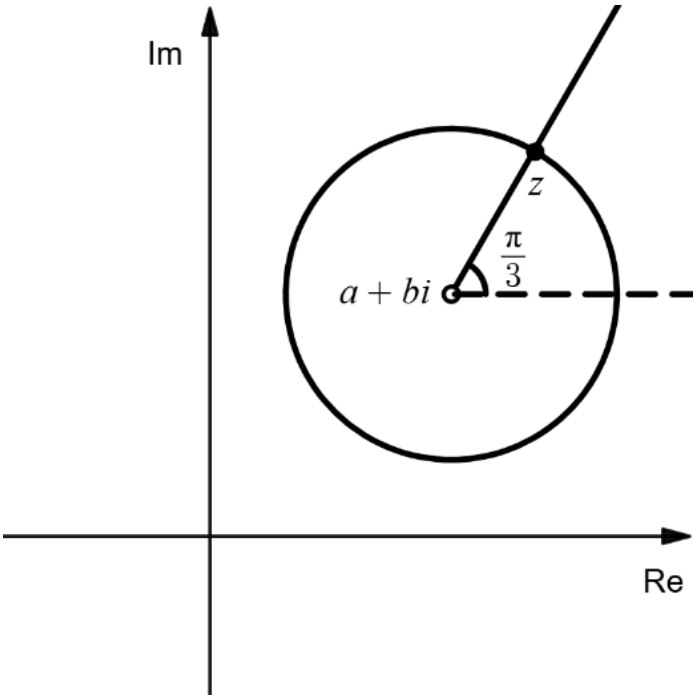
So $x=2+2\sqrt{2}$ and $y=-2\sqrt{2}$

$$\therefore z=2+2\sqrt{2}-2\sqrt{2}i$$

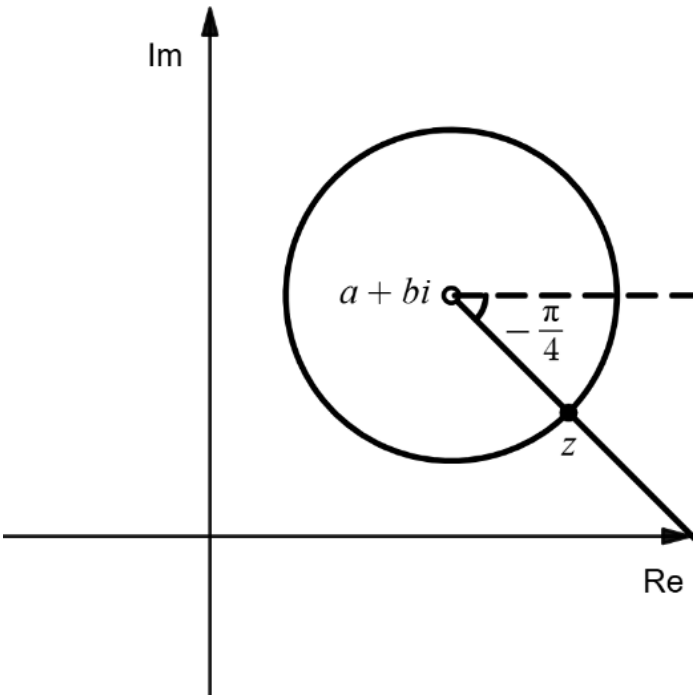
29 c)

$$(2+2\sqrt{2})^2+(-2\sqrt{2})^2=20+8\sqrt{2}$$
$$|z|^2=20+8\sqrt{2}$$

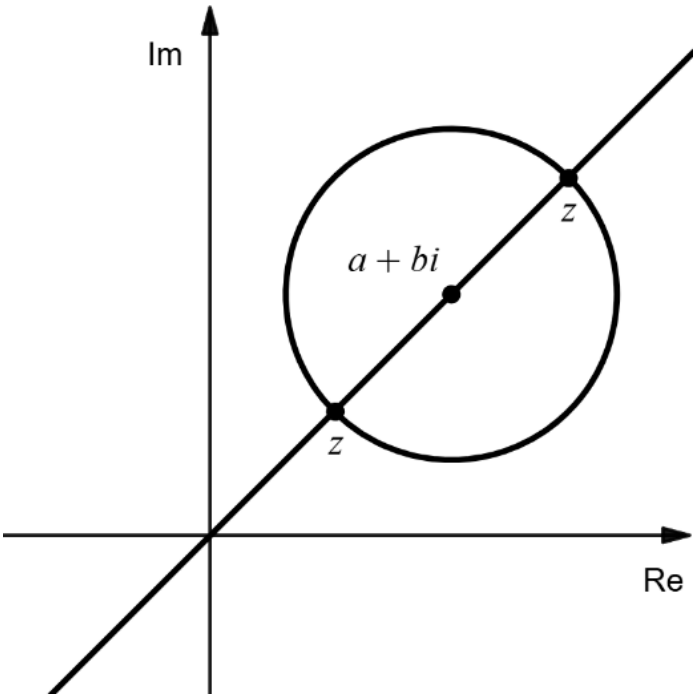
30 a)



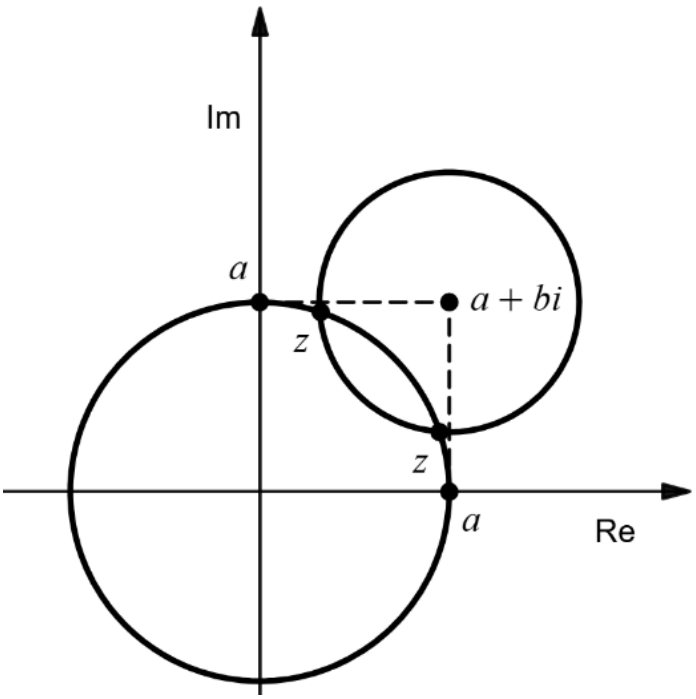
30 b)



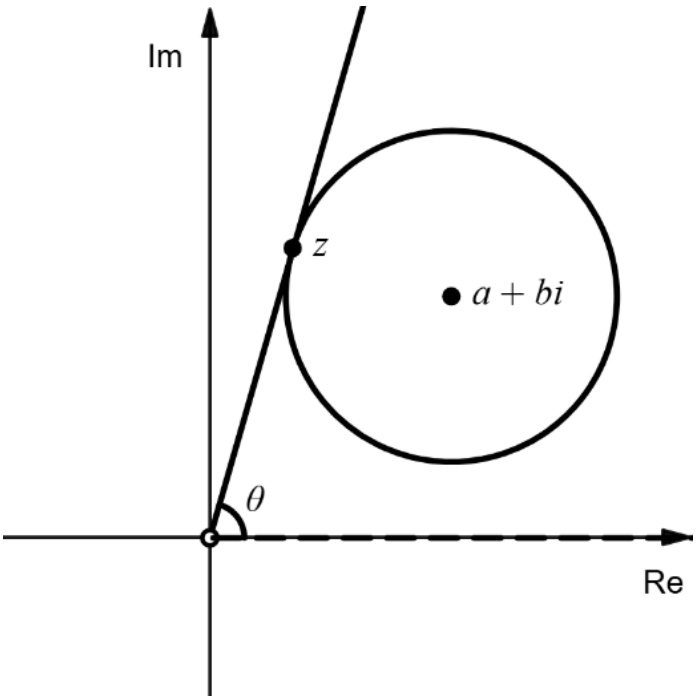
30 c)



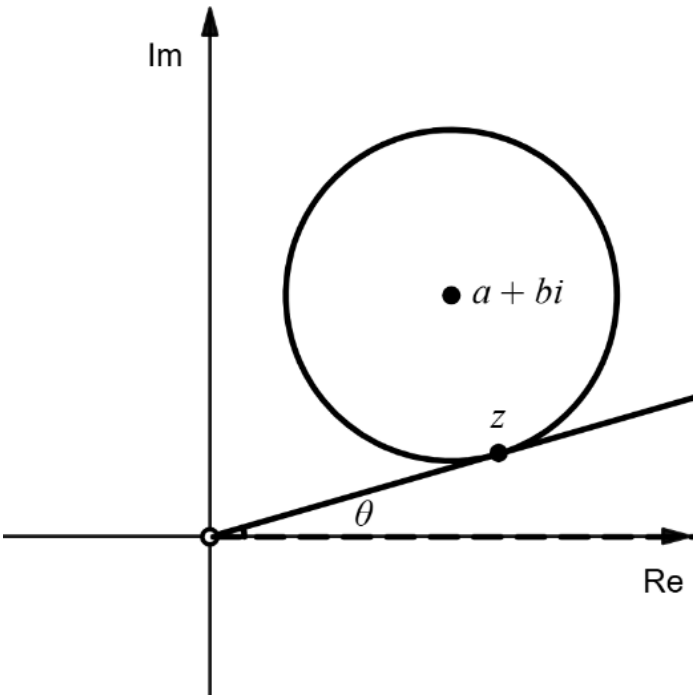
30 d)



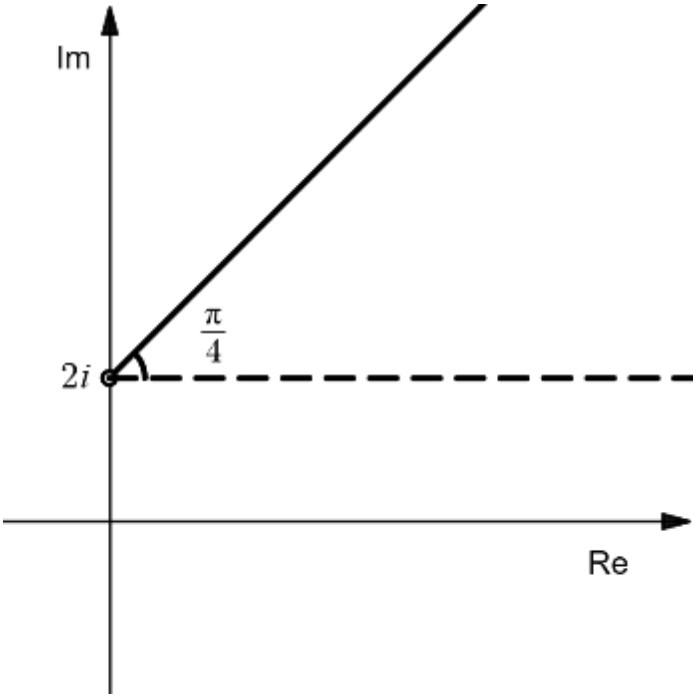
30 e)



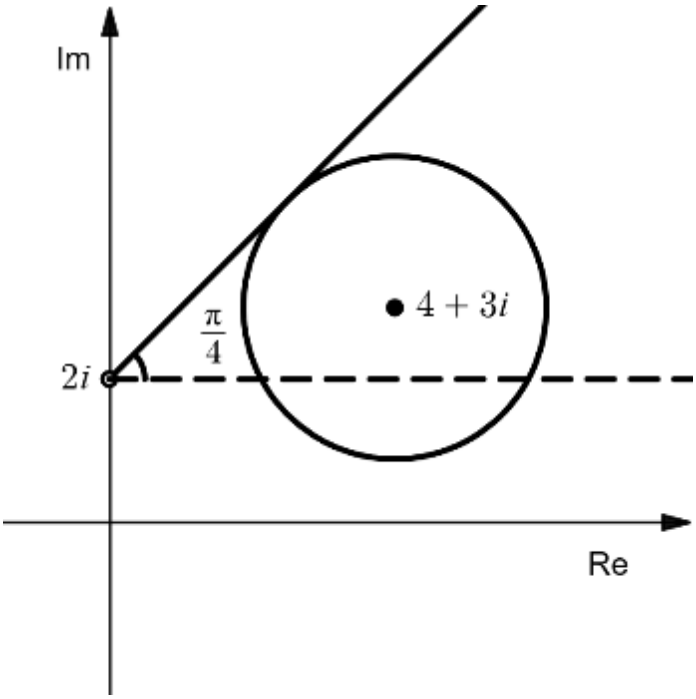
30 f)



31 a)



31 b)



Need the radius for which the line touches the circle.

$(x-4)^2+(y-3)^2=k^2$ and $y=x+2$

$$(x-4)^2+(x+2-3)^2=k^2$$
$$(x-4)^2+(x-1)^2=k^2$$
$$x^2-8x+16+x^2-2x+1-k^2=0$$
$$2x^2-10x+17-k^2=0$$

Consider $b^2-4ac=0$:

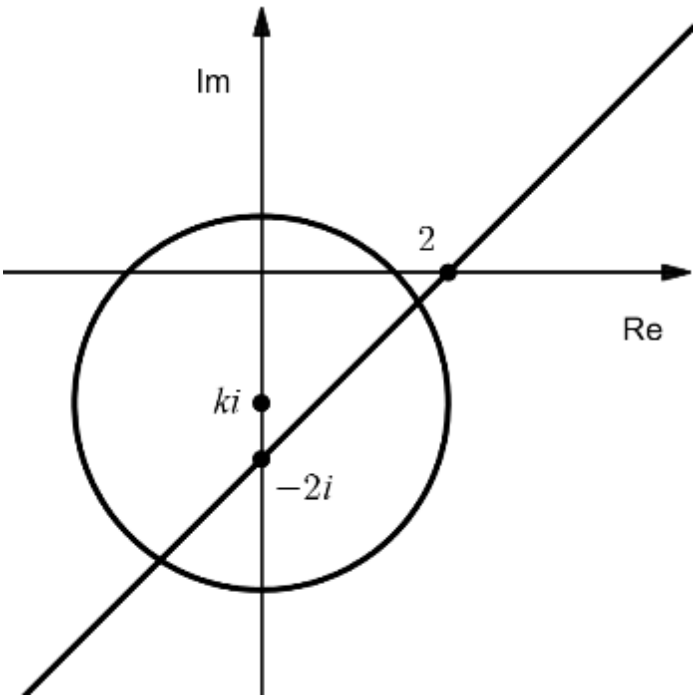
$$(-10)^2-4\times 2\times(17-k^2)=0$$
$$8k^2-36=0$$
$$k^2=\frac{9}{2}\Rightarrow k=\pm\frac{3}{\sqrt{2}}=\pm\frac{3\sqrt{2}}{2}$$

Since k is positive, $k=\frac{3\sqrt{2}}{2}$

$$\therefore |z-4-3i|=\frac{3\sqrt{2}}{2}$$

32)

As per the diagram:



There must be solution(s) to $x^2 + (y-k)^2 = 4$ and $y = x - 2$

$$x^2 + (x - 2 - k)^2 = 4$$
$$x^2 + x^2 - 4x - 2kx + k^2 + 4k + 4 = 4$$
$$2x^2 - (4 + 2k)x + (4k + k^2) = 0$$

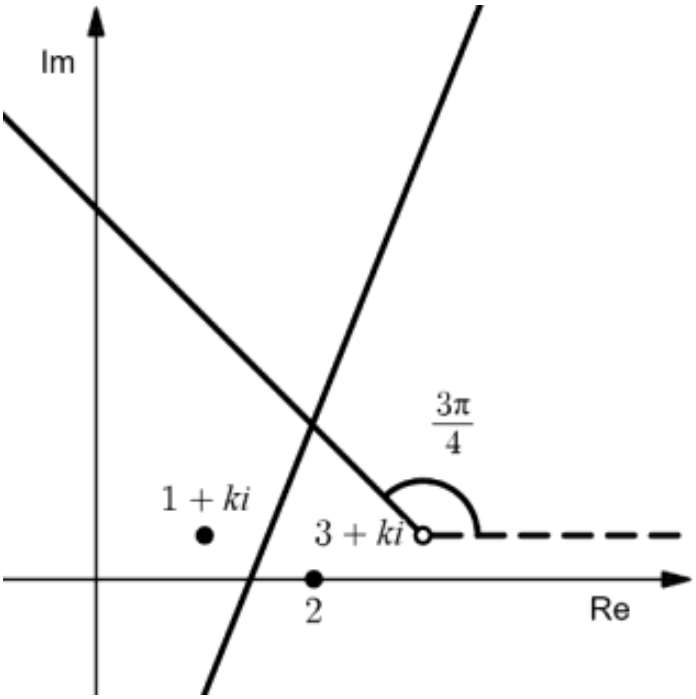
Consider $b^2 - 4ac = 0$:

$$(4 + 2k)^2 - 4 \times 2(4k + k^2) = 0$$
$$16 + 16k + 4k^2 - 32k - 8k^2 = 0$$

$k^2 + 4k - 4 = 0$ giving $k = -2 \pm 2\sqrt{2}$

$$\therefore -2 - 2\sqrt{2} \leq k \leq -2 + 2\sqrt{2}$$

33)



Midpoint: $\left(\frac{3}{2}, \frac{k}{2}\right)$ and gradient: $-\frac{k}{1}$

$$y - \frac{k}{2} = \frac{1}{k}\left(x - \frac{3}{2}\right) \Rightarrow y = \frac{1}{k}x - \frac{3}{2k} + \frac{k}{2}$$

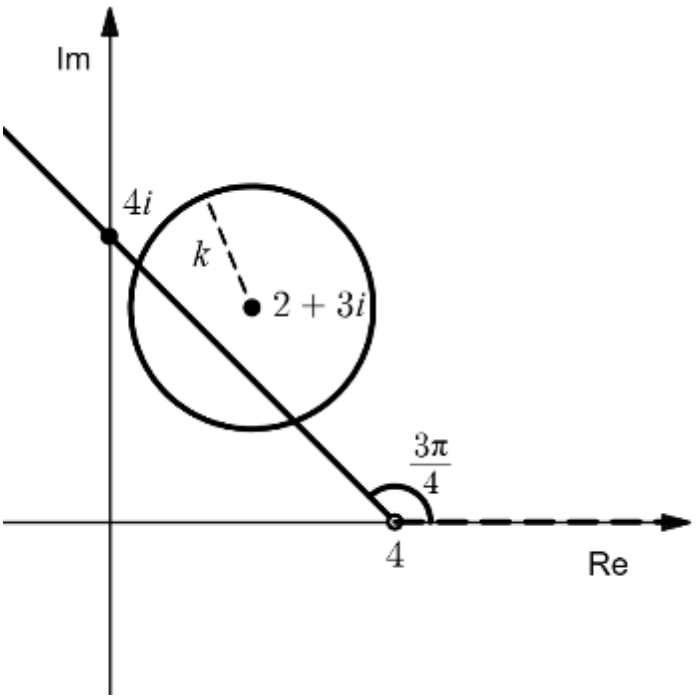
Since k is positive, the gradient of the line is positive.

We need the point where the line passes through the point $3 + ki$

$$k = \frac{1}{k} \times 3 - \frac{3}{2k} + \frac{k}{2}$$
$$k = \frac{6}{2k} - \frac{3}{2k} + \frac{k^2}{2k}$$
$$2k^2 = 3 + k^2$$
$$k^2 = 3 \therefore k = \sqrt{3}$$

So that the interval is $0 < k < \sqrt{3}$

34)



Need the values of k for which the line intersects with the circle.

$(x-2)^2+(y-3)^2=k^2$ and $y=-x+4$

$$(x-2)^2+(-x+4-3)^2=k^2$$
$$(x-2)^2+(-x+1)^2=k^2$$
$$x^2-4x+4+x^2-2x+1=k^2$$
$$2x^2-6x+5-k^2=0$$

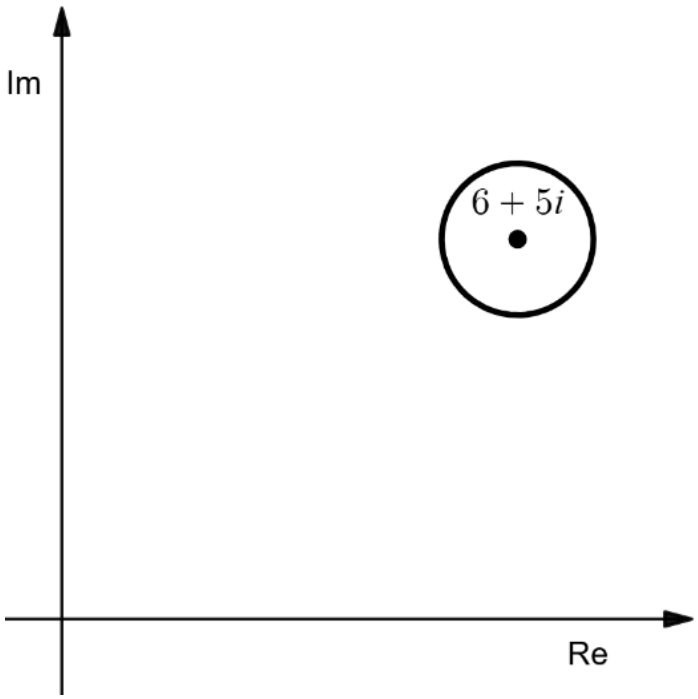
Consider $b^2-4ac=0$:

$$(-6)^2-4\times 2\times(5-k^2)=0$$
$$36-40+8k^2=0$$
$$k^2=\frac{1}{2}\Rightarrow k=\pm\sqrt{\frac{1}{2}}=\pm\frac{\sqrt{2}}{2}$$

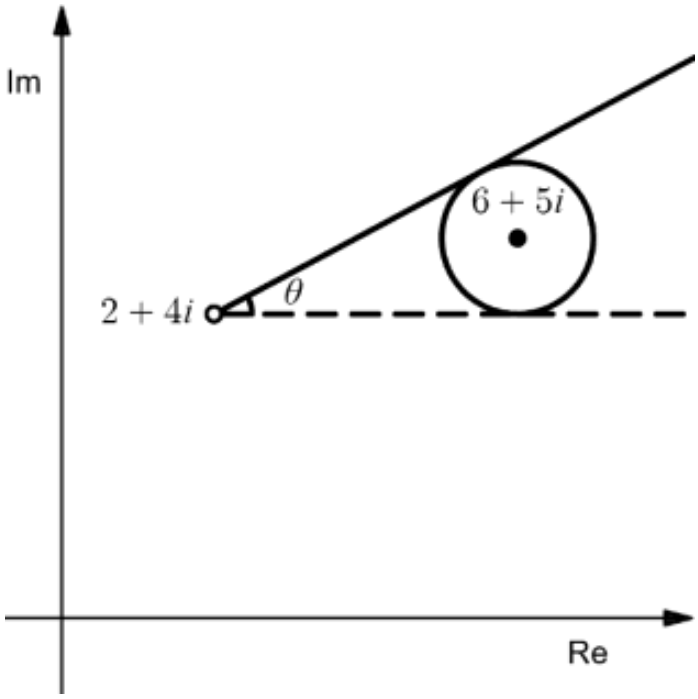
Since k is positive, $k=\frac{\sqrt{2}}{2}$

$$\therefore k\geq\frac{\sqrt{2}}{2}$$

35 a)



35 b)



Need the values of θ for which the line intersects with the circle exactly once.

$(x-6)^2+(y-5)^2=1$ and $y-4=m(x-2)$ where $m=\tan\theta$

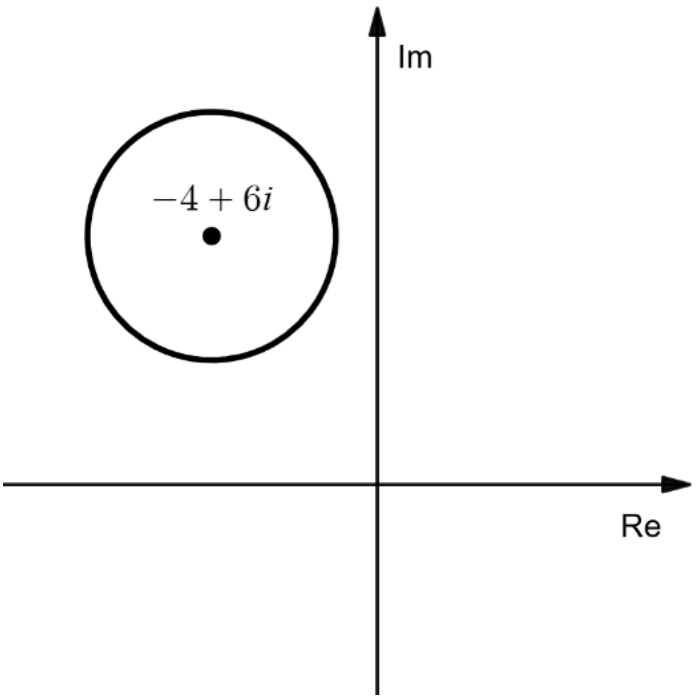
$$(x-6)^2+(mx-2m+4-5)^2=1$$
$$(x-6)^2+(mx-2m-1)^2=1$$
$$x^2-12x+36+m^2x^2-4m^2x+4m^2-2mx+4m+1=1$$
$$(1+m^2)x^2+(-4m^2-2m-12)x+(4m^2+4m+36)=0$$

Consider $b^2 - 4ac = 0$:

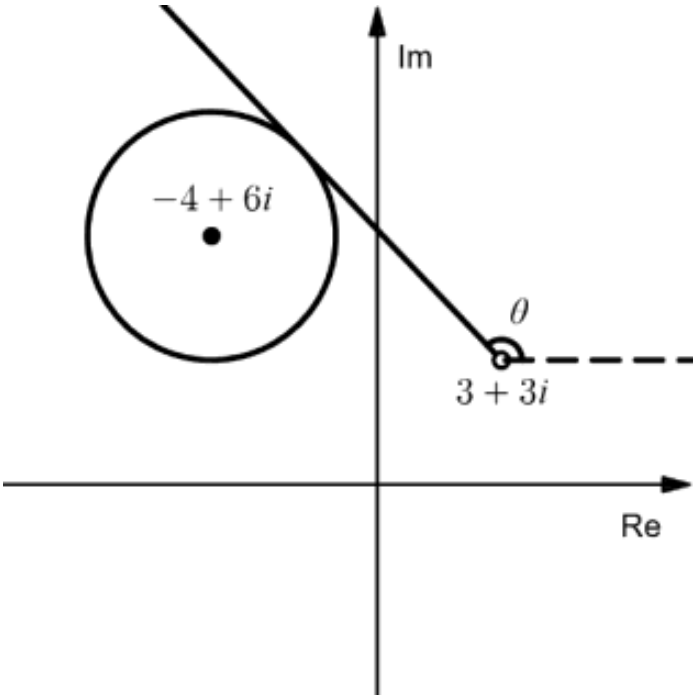
$$(-4m^2 - 2m - 12)^2 - 4(1 + m^2)(4m^2 + 4m + 36) = 0$$
$$16m^4 + 16m^3 + 100m^2 + 48m + 144$$
$$- 16m^4 - 16m^3 - 160m^2 - 16m - 144 = 0$$
$$32m - 60m^2 = 0$$
$$m(8m - 15) = 0$$
$$\therefore m = 0 \text{ or } m = \frac{15}{8}$$

$\tan\theta = \frac{8}{15} \therefore \theta = \arctan\left(\frac{8}{15}\right)$ as req.

36 a)



36 b)



Need the values of θ for which the line intersects with the circle exactly once.
$(x+4)^2+(y-6)^2=9$ and $y-3=m(x-3)$

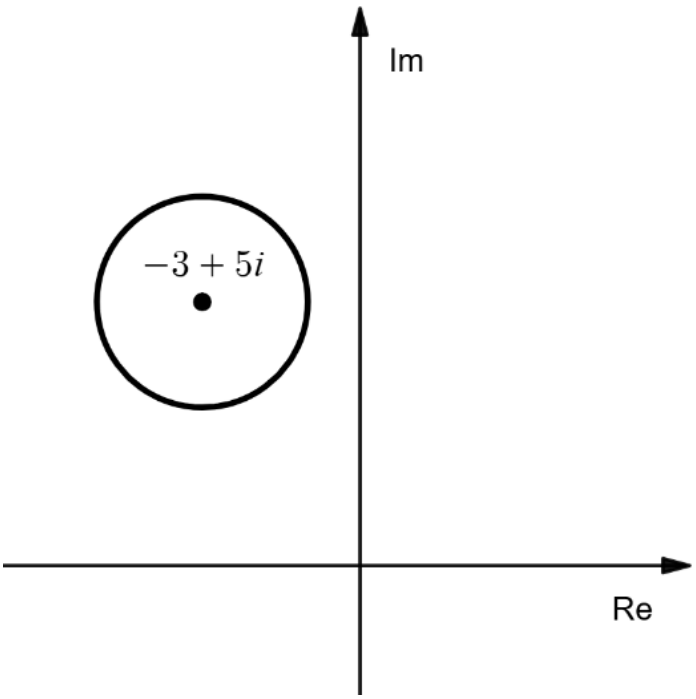
$$(x+4)^2+(mx-3m+3-6)^2=9$$
$$(x+4)^2+(mx-3m-3)^2=9$$
$$x^2+8x+16+m^2x^2-6m^2x+9m^2-6mx+18m+9=9$$
$$(1+m^2)x^2+(8-6m^2-6m)x+(9m^2+18m+16)=0$$

Consider $b^2-4ac=0$:

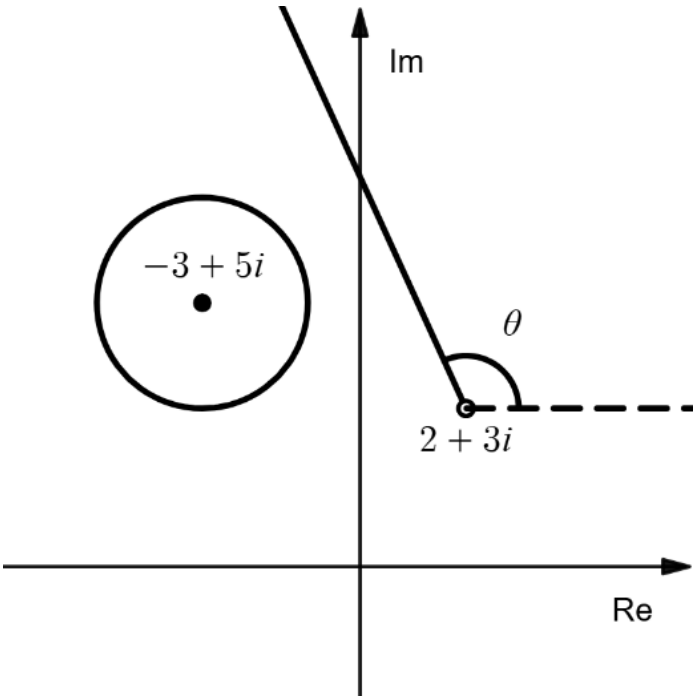
$$(8-6m^2-6m)^2-4(1+m^2)(9m^2+18m+16)=0$$
$$36m^4+72m^3-60m^2-96m+64$$
$$-36m^4-72m^3-100m^2-72m-64=0$$
$$-160m^2-168m=0$$
$$m(20m+21)=0$$
$$\therefore m=0 \text{ or } m=-\frac{21}{20}$$

Which gives $\arctan\left(\frac{20}{21}\right)$ $\therefore \theta=\frac{\pi}{2}+\arctan\left(\frac{20}{21}\right)$ as req.

37 a)



37 b)



Need the values of θ for which the line intersects with the circle.

$(x+3)^2+(y-5)^2=4$ and $y-3=m(x-2) \Rightarrow y=mx-2m+3$

$$(x+3)^2+(mx-2m+3-5)^2=4$$
$$(x+3)^2+(mx-2m-2)^2=4$$
$$x^2+6x+9+m^2x^2-4m^2x-4mx+4m^2+8m+4=4$$
$$(1+m^2)x^2+(6-4m-4m^2)x+(4m^2+8m+9)=0$$

Consider $b^2 - 4ac = 0$:

$$(6 - 4m - 4m^2)^2 - 4(1 + m^2)(4m^2 + 8m + 9) = 0$$
$$16m^4 + 32m^3 - 32m^2 - 48m + 36$$
$$- 16m^2 - 32m - 36 - 16m^4 - 32m^3 - 36m^2 = 0$$
$$84m^2 + 80m = 0$$
$$m(21m + 20) = 0$$

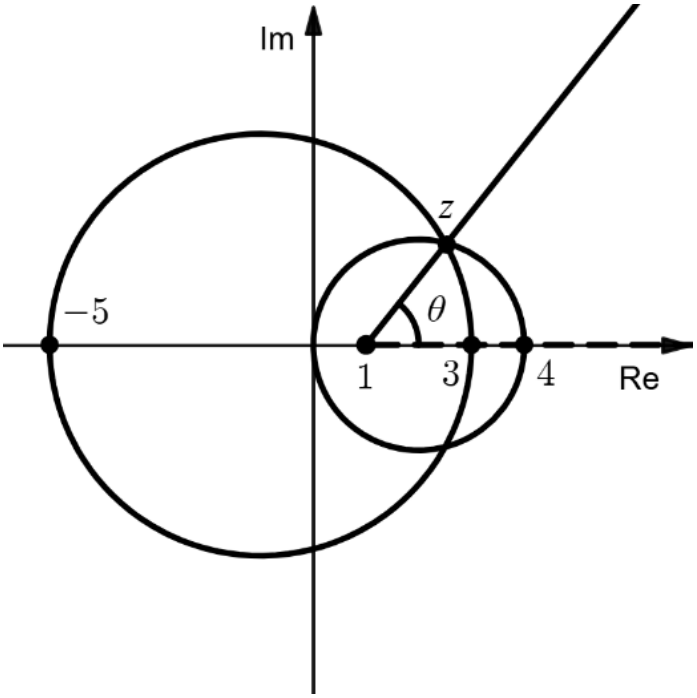
$\therefore m = 0$ or $m = -\frac{20}{21}$ so $\tan\phi = 0$ or $\tan\phi = -\frac{20}{21}$ where $\theta + \phi = \pi$

So $\theta = \pi - \tan\phi = \pi$ or $\theta = \pi - \arctan\left(\frac{20}{21}\right) = 2.38$ or $\theta = \frac{\pi}{2} + \arctan\left(\frac{21}{20}\right) = 2.38$

$$\therefore \frac{\pi}{2} + \arctan\left(\frac{21}{20}\right) \leq \theta \leq \pi \text{ or } 2.38 \leq \theta \leq 3.14$$

38 a)

There are two values of z which satisfy the loci, one of which is shown in the diagram below:



Solve $(x-2)^2 + y^2 = 4$ and $(x+1)^2 + y^2 = 16$ simultaneously:

$$(x-2)^2 + 16 - (x+1)^2 = 4$$
$$x^2 - 4x + 4 + 16 - x^2 - 2x - 1 - 4 = 0$$
$$-6x + 15 = 0 \therefore x = \frac{5}{2} \text{ so } y = \pm\frac{\sqrt{15}}{2}$$

$z = \frac{5}{2} + \frac{\sqrt{15}}{2}i$ and $z = \frac{5}{2} - \frac{\sqrt{15}}{2}i$

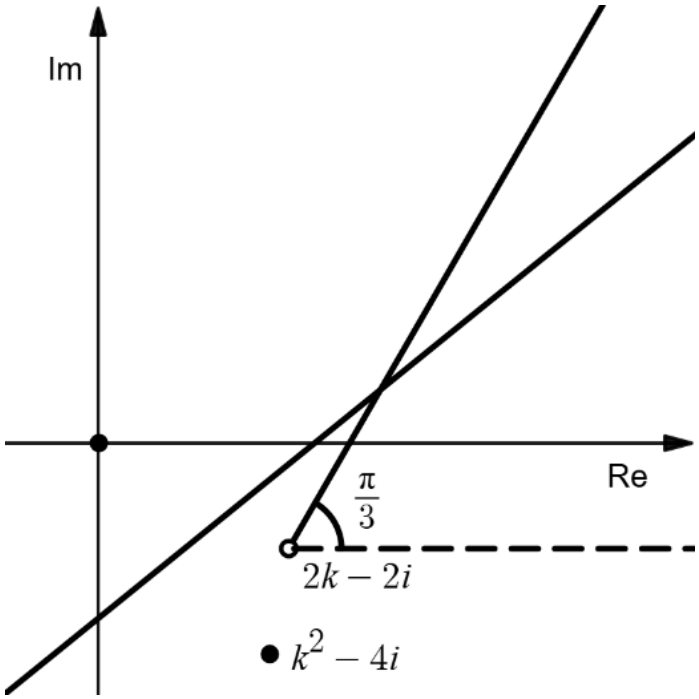
$$\arctan\left(\frac{\frac{\sqrt{15}}{2}}{\frac{5}{2} - 1}\right) = \arctan\left(\frac{\sqrt{15}}{3}\right) = 0.912 \text{ (3 SF)}$$

$\therefore \theta = 0.912$ or $\theta = -0.912$ (3 SF)

38 b)

$$-0.912 < \theta < 0.912$$

39 a)



Midpoint: $\left(\frac{1}{2}k^2, -2\right)$ and gradient: $m = -\frac{4}{k^2}$

$$y + 2 = \frac{k^2}{4}\left(x - \frac{1}{2}k^2\right) \Rightarrow y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$$

Half-line:

$$y + 2 = \tan\frac{\pi}{3}(x - 2k) \Rightarrow y = \sqrt{3}x - 2\sqrt{3}k - 2$$

Solve $y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$ and $y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$ simultaneously:

$$\frac{k^2}{4}x - \frac{k^4}{8} - 2 = \sqrt{3}x - 2\sqrt{3}k - 2$$
$$\frac{k^2}{4}x - \sqrt{3}x = -2\sqrt{3}k + \frac{k^4}{8}$$
$$\left(\frac{k^2}{4} - \sqrt{3}\right)x = -2\sqrt{3}k + \frac{k^4}{8}$$
$$(2k^2 - 8\sqrt{3})x = -16\sqrt{3}k + k^4$$
$$x = \frac{k^4 - 16\sqrt{3}k}{2k^2 - 8\sqrt{3}} \text{ as req}$$

39 b)

$\text{Im}(z) = 0$ when $y = 0$

$$0 = \sqrt{3}x - 2\sqrt{3}k - 2 \Rightarrow x = \frac{2\sqrt{3}}{3} + 2k$$

and $0 = \frac{k^2}{4}x - \frac{k^4}{8} - 2 \Rightarrow 0 = \frac{k^2}{4}\left(\frac{2\sqrt{3}}{3} + 2k\right) - \frac{k^4}{8} - 2$

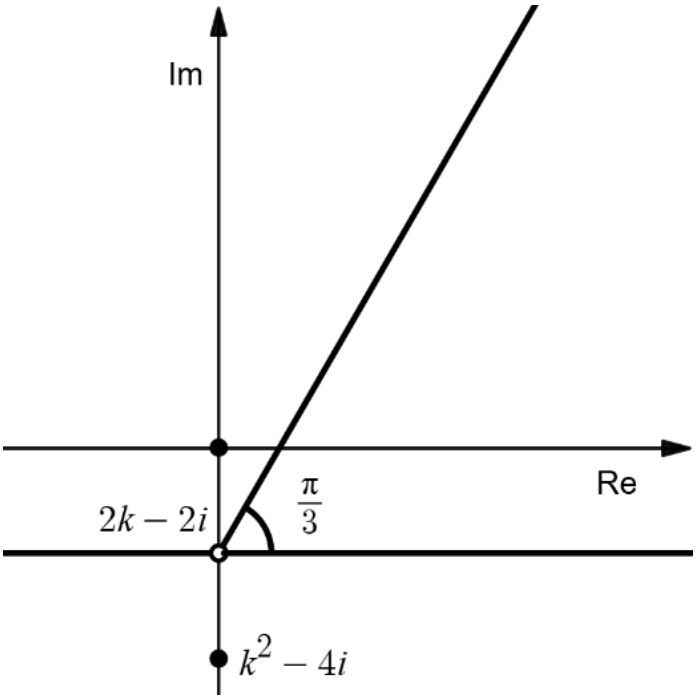
$$0 = \frac{\sqrt{3}}{6}k^2 + \frac{1}{2}k^3 - \frac{k^4}{8} - 2$$

Solving gives $k = 1.6118\ldots$ and $k = 4.3363\ldots$

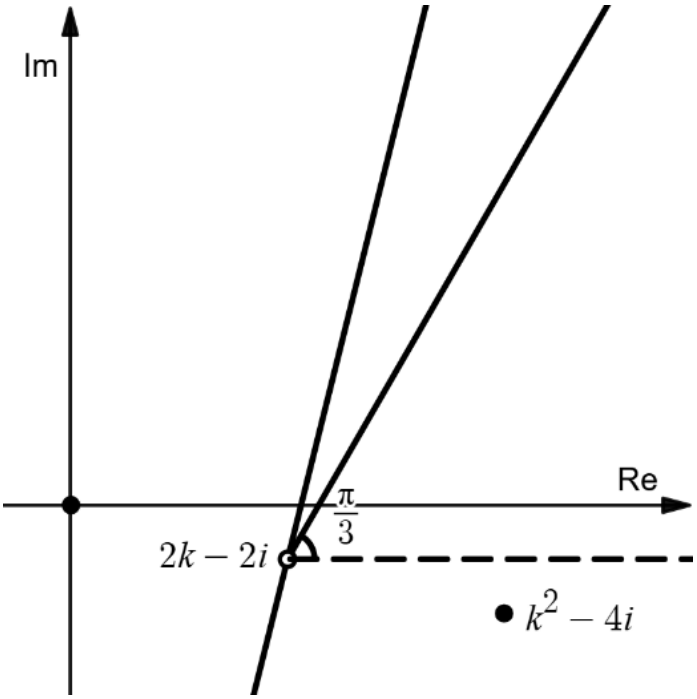
This gives $x = 4.38$ and $x = 9.83$

$\therefore \text{Re}(z) = 4.38$ when $k = 1.61$ and $\text{Re}(z) = 9.83$ when $k = 4.34$

39 c)



When $k = 0$, $\text{Re}(z) = 0$ so the half-line starts at $-2i$ which means this point is not included.



When $k = 4$, the midpoint between 0 and $k^2 - 4i$ is $8 - 2i$ which is the start point for the half-line which means this point is not included.

39 d)

The two lines are parallel when $\frac{k^2}{4} = \sqrt{3}$ so when $k = \pm 2 \times 3^{\frac{1}{4}}$

Consider k as it varies between $-2 \times 3^{\frac{1}{4}}$, 0, $2 \times 3^{\frac{1}{4}}$ and 4

$$\therefore k < -2 \times 3^{\frac{1}{4}}, \quad 0 < k < 2 \times 3^{\frac{1}{4}}, \quad k > 4$$

40 a)

Let $z = x + yi$

$$\begin{aligned}\frac{z-4}{z} &= \frac{x+yi-4}{x+yi} \times \frac{x-yi}{x-yi} \\ &= \frac{x^2 + xyi - 4x - xyi + y^2 + 4yi}{x^2+y^2} \\ &= \frac{x^2 - 4x + y^2 + 4yi}{x^2+y^2}\end{aligned}$$

$$\therefore \frac{z-4}{z} = \frac{x^2-4x+y^2}{x^2+y^2} + \frac{4y}{x^2+y^2}i \text{ as req.}$$

40 b)

Since $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$ then $\tan\left(\frac{\pi}{3}\right) = \frac{\frac{4y}{x^2+y^2}}{\frac{x^2-4x+y^2}{x^2+y^2}} = \frac{4y}{x^2-4x+y^2}$

So $\sqrt{3} = \frac{4y}{x^2-4x+y^2}$

$$\sqrt{3}x^2 - 4\sqrt{3}x + \sqrt{3}y^2 = 4y$$

$$\sqrt{3}x^2 - 4\sqrt{3}x + \sqrt{3}y^2 - 4y = 0$$

$$\sqrt{3}(x-2)^2 - 4\sqrt{3} + \sqrt{3}\left(y - \frac{2\sqrt{3}}{3}\right)^2 - \frac{4\sqrt{3}}{3} = 0$$

$$(x-2)^2 - 4 + \left(y - \frac{2\sqrt{3}}{3}\right)^2 - \frac{4}{3} = 0$$

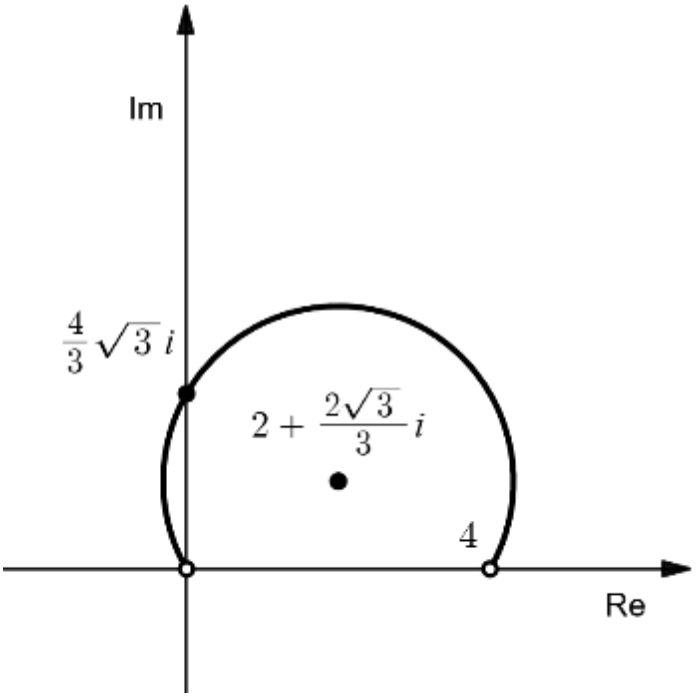
$(x-2)^2 + \left(y - \frac{2\sqrt{3}}{3}\right)^2 = \frac{16}{3}$ which is an equation of a circle

The circle crosses the real axis at $z = 0$ and $z = 4$

Since the argument is positive, only z must lie above the real axis where $z = 0$ and $z = 4$ have no solutions to $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$

It is the circle for which only non-zero positive imaginary numbers are included.

40 c)


Im
$\frac{4}{3}\sqrt{3}\,i$
$2+\frac{2\sqrt{3}}{3}i$
4
Re